GENERAL SCIENCE GRADE 10: THE CELL

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 1.0: Life Science
Sub-Strand	Sub-Strand 1.2: The Cell
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Types of microscopes (light microscope and electron microscope) – Differences between light and electron microscopes – Cell structure and functions as seen under the electron microscope – Structure of the plant cell under the electron microscope – Structure of the animal cell under the electron microscope – Comparison of plant and animal cell structures under light and electron microscopes – Functions of the components of plant and animal cells – Levels of cell organisation (organelles → cells → tissues → organs → organ systems → organisms) – Project: constructing a model of a plant and/or animal cell using locally available materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the differences between a light and an electron microscope, b) describe the plant and animal cell under an electron microscope, c) compare plant and animal cell structures under light and electron microscopes, d) explain the functions of the components of plant and animal cells, e) construct a concept map of the levels of cell organisation, f) appreciate the function of the cell as a basic unit of life.
Core Competencies	– Creativity and Imagination: as the learner constructs a model of plant and animal cells as seen under the electron microscope using locally available materials such as maize grains, bean seeds, bottle tops, and polythene bags sourced from Nairobi's Gikomba market. – Digital Literacy: as the learner searches for information from the internet and digital media on the structure of a cell as seen under the electron microscope, including photomicrographs of onion epidermal cells and cheek cells. – Learning to Learn: as the learner reflects on the project work of model-building and adjusts the cell model iteratively based on peer and teacher feedback.
Core Values	– Respect: as the learner considers others' opinions while working on practical activities in groups such as construction of cell models, ensuring every group member's contribution is acknowledged. – Social Justice: as the learner seeks equity, equality and gender consideration in distribution of learning resources such as

	sharing of locally available materials for making cell models.
Science & Engineering Practices	– Developing and Using Models: as the learner draws, labels and physically constructs models of plant and animal cells, revising the model across lessons to incorporate new understanding of organelle functions and levels of organisation. – Analysing and Interpreting Data: as the learner observes photomicrographs of plant cells (e.g., cells from a succulent leaf found along Lake Victoria's shores) and animal cells, identifying organelles and recording comparative observations in a data table. – Asking Questions and Defining Problems: as the learner generates questions on the Driving Question Board about why plant cells from ugali maize kernels have cell walls while animal cheek cells do not, driving further investigation. – Constructing Explanations: as the learner uses evidence from photomicrograph observations and model-building to explain why the cell is considered the basic unit of life and how levels of organisation lead to complex organisms such as a Maasai giraffe or a maize plant. – Obtaining, Evaluating and Communicating Information: as the learner researches electron microscope images using digital devices, evaluates sources, and presents findings using labelled diagrams and digital presentations shared with peers.
Pertinent & Contemporary Issues (PCIs)	– Environmental Conservation: as the learner manages wastes responsibly to avoid pollution while preparing plant and animal cell models, disposing of organic materials (e.g., onion peels, maize husks) appropriately in school compost pits. – Safety and Security: as the learner observes safety precautions when preparing the models, including safe handling of sharp tools (scalpels, needles) used to section plant specimens such as sukuma wiki leaves.
Career Connections	– Medical Laboratory Scientist: this sub-strand develops skills in microscopy and cell identification directly applied in hospital labs such as Kenyatta National Hospital where blood and tissue cells are routinely examined. – Biomedical Research Scientist: understanding cell ultrastructure is foundational for researchers at institutions like KEMRI (Kenya Medical Research Institute) working on cell-level disease mechanisms. – Histotechnologist / Pathologist: knowledge of animal and plant cell structures and tissue organisation underpins the preparation and examination of tissue sections in pathology departments across Kenya. – Agricultural Biotechnologist: understanding plant cell structure (cell wall, chloroplasts, vacuole) is prerequisite knowledge for tissue culture work done at institutions like KARI (Kenya Agricultural and Livestock Research Organization) to improve crops such as maize and tea in the Rift Valley. – Secondary School Science Teacher: the ability to explain cell structure, use microscopes, and construct concept maps on levels of organisation is a core professional skill for TSC-registered science teachers across Kenya.
Focus for Lessons	This unit anchors on the cell as the fundamental unit of life, guiding Grade 10 learners to move from the familiar light microscope used in Junior School Integrated Science to the higher-resolution world of the electron microscope. Learners build understanding progressively — from comparing microscope types, to examining cell ultrastructure, to explaining organelle functions, to mapping the hierarchy of organisation from organelle to organism — using Kenyan biological specimens (sukuma wiki leaves, onion epidermal cells, cheek cells) as concrete reference points. The unit culminates in a project where learners use locally available materials to construct three-dimensional cell models, consolidating conceptual understanding through creative, hands-on making.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive? KICD KEY INQUIRY QUESTION: 1. What makes up an organism?
Anchoring Phenomenon	A Grade 10 learner at a school in Kisumu notices that when a piece of sukuma wiki (kale) leaf is left in the sun for several days, it wilts and eventually dries and dies — yet if you look at a thin slice of a fresh sukuma wiki leaf under the school's light microscope,

	you can see tiny box-like compartments packed with green granules. The learner wonders: what are these compartments, what is inside them, and how does what is inside them keep the leaf alive? When the teacher shows a photomicrograph of the same leaf taken with an electron microscope, far more internal detail is visible — folded membranes, dark oval bodies, and a large fluid-filled space. The learner is puzzled: the light microscope and the electron microscope seem to reveal completely different levels of detail in what is supposedly the same cell. Why do the two microscopes show such different pictures? What are all those internal structures, and what are they each doing to keep the sukuma wiki plant — and ultimately the learner who eats it — alive?
Storyline Thread	Lesson 1 — Predict & Anchor Phenomenon (Microscopes as Windows): Learners observe a fresh sukuma wiki leaf slice under the school light microscope and record what they see. Teacher reveals an electron micrograph of the same leaf cell. Learners predict: what extra structures might the electron microscope reveal and why? Class constructs an initial Driving Question Board (DQB) capturing "what we think we know" and "what we wonder." Focus: eliciting prior knowledge and generating questions about cell structure. Lesson 2 — Observe Phase (Light vs. Electron Microscope): Learners use print and digital media to research the differences between light and electron microscopes (resolution, magnification, specimen preparation, cost). Pairs create a comparison table. Teacher cold-calls 4 pairs to share one key difference each. Focus: SLO (a) — explaining differences between light and electron microscopes; SEP: Obtaining and Communicating Information. Lesson 3 — Observe Phase (Animal Cell Ultrastructure): Learners observe and annotate electron micrograph photomicrographs of animal cells (e.g., human cheek cell). In groups of 3, learners identify visible organelles and record in a labelled diagram. Teacher-guided class discussion connects each organelle to a "job" using Kenyan analogies (mitochondria = jiko/charcoal stove providing energy; nucleus = county governor's office holding instructions). DQB updated. Focus: SLO (b) — animal cell under electron microscope. Lesson 4 — Observe Phase (Plant Cell Ultrastructure & Comparison): Learners observe electron micrograph photomicrographs of plant cells (onion, sukuma wiki). Groups add a plant cell column to their diagram table and identify structures present in plant cells but absent in animal cells (cell wall, chloroplasts, large central vacuole). Teacher facilitates a Venn-diagram whole-class comparison. Focus: SLO (b) and SLO (c) — plant cell structure and comparison with animal cell. Lesson 5 — Explain Phase (Organelle Functions): Learners receive a "functions card sort" — cards describing organelle functions matched to organelle name cards. Pairs complete the sort, then groups of 4 discuss disagreements. Teacher reveals correct matches and connects each function back to the anchoring phenomenon (e.g., chloroplasts in sukuma wiki cells carry out photosynthesis that produces the food the learner eats). DQB: learners add answers to earlier questions. Focus: SLO (d) — functions of components of plant and animal cells. Lesson 6 — Explain Phase (Levels of Cell Organisation): Learners construct a concept map linking organelle → cell → tissue → organ → organ system → organism, using two Kenyan examples: (i) the sukuma wiki plant (leaf cell → leaf tissue → leaf → shoot system → plant) and (ii) the human body (muscle cell → muscle tissue → stomach → digestive system → human). Groups present their concept maps; teacher facilitates peer feedback. Focus: SLO (e) — concept map of levels of cell organisation; SEP: Developing and Using Models. Lesson 7 — Model Building Phase (3D Cell Model Project): Learners use locally available materials (bottle tops, polythene bags, maize grains, bean seeds, clay, string, cardboard) to construct a 3D model of either a plant or animal cell, labelling each organelle and writing its function on a tag. Teacher circulates, asking each learner to justify one modelling choice. DQB revisited: learners identify which questions have now been answered by the model. Focus: SLO (f) and Creativity & Imagination core competency;

	SEP: Developing and Using Models. Lesson 8 — Driving Question Board Synthesis & Assessment: Groups display their models. Gallery walk — each group leaves sticky-note feedback on two other models. Class returns to the original anchoring phenomenon (sukuma wiki leaf wilting) and collectively constructs a full written explanation linking cell structure, organelle function, and levels of organisation to why the leaf stays alive or dies. Teacher administers a short performance task: learners draw and label a plant cell from a photomicrograph and write a paragraph explaining how its structures support life. DQB: remaining unanswered questions flagged as "leads into Sub-Strand 1.3 Nutrition in Animals." Focus: synthesis, assessment, and transition to next sub-strand.
Supporting Phenomena	– A farmer in Kericho observes that tea leaves affected by a fungal blight turn brown and collapse — at the cellular level, the cell walls and membranes of the leaf cells have been destroyed; learners examine healthy vs. diseased leaf photomicrographs to see what cellular structures are lost. – A student at a Nairobi school cuts their finger and notices the wound heals over several days — this is puzzling because the skin seems solid; the phenomenon prompts inquiry into how animal cells (skin cells lacking cell walls) can multiply and repair, linking to cell structure and the absence of a rigid cell wall in animal cells. – A maize (corn) grain soaked in water overnight swells noticeably; learners observe thin cross-sections under the light microscope and see that individual cells have increased in size — this anchors inquiry into the large central vacuole of plant cells and its role in water uptake. – A drop of human blood smeared on a slide and stained with methylene blue reveals flat, disc-shaped red blood cells and larger, irregularly shaped white blood cells — the same organism contains cells of very different shapes and internal organisation, raising the question of how one organism can have so many different cell types (connecting to levels of cell organisation and tissue specialisation).

LESSON 1 (80 minutes (2 × 40-minute lessons)): Predict & Anchor Phenomenon — Microscopes as Windows into the Cell	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor the unit in the sukuma wiki wilting phenomenon and elicit learners' prior knowledge about cell structure by comparing what is visible under a light microscope versus an electron microscope, thereby launching the Driving Question Board.
Knowledge	Learners will know that: (a) cells are the basic structural units of living organisms such as sukuma wiki and the human body; (b) light microscopes and electron microscopes reveal different levels of cellular detail; (c) plant cells contain visible internal compartments and granules (chloroplasts) observable under a light microscope, and additional structures (folded membranes, oval bodies, large fluid-filled spaces) visible only under an electron microscope.
Skills	Learners will be able to: (a) prepare and observe a wet mount of a thin sukuma wiki leaf section under a light microscope; (b) record and annotate observational drawings from both a light microscope and an electron micrograph; (c) formulate evidence-based predictions about structures not yet explained; (d) construct an initial Driving Question Board (DQB) that distinguishes 'what we think we know' from 'what we wonder.'
Attitudes	Learners will appreciate that careful observation and honest acknowledgement of uncertainty are the starting points of scientific inquiry; they will respect peers' differing initial ideas and value the role of improving technology (from light to electron microscopy) in expanding human understanding of life.
Key Inquiry Question	What makes up an organism?
Purpose in Storyline	This is the opening lesson of the unit. It anchors the entire storyline in the observable phenomenon of sukuma wiki wilting and dying, then zooms learners into the microscopic world to reveal complexity they cannot yet explain. By surfacing prior knowledge and generating genuine questions on the DQB, the lesson creates the intellectual need that will drive all subsequent lessons on cell organelles, their functions, and levels of organisation.
Safety Notes	– Handle glass slides and cover slips carefully — report any breakage immediately to the teacher. • Use only the blade provided by the teacher to cut the sukuma wiki leaf; cut away from fingers on a cutting board. • Do not touch the microscope lamp — it becomes hot during use. • Wash hands with soap and water after handling plant material. • Dispose of used slides in the designated sharps container, not the general waste bin.

B. LESSON OVERVIEW

Learners arrive at a class where a fresh sukuma wiki leaf and a wilted, dried leaf are displayed side by side — the anchoring phenomenon. After briefly discussing what they notice, pairs prepare wet-mount slides of a thin section of the fresh leaf and observe it under the school's light microscope, recording annotated drawings. The teacher then reveals a projected electron micrograph of the same leaf cell. Learners compare the two images, identify structures they cannot yet explain, and write individual predictions about what the extra structures do. The lesson closes with the class co-constructing an initial Driving Question Board (DQB), populating it with 'what we think we know' and 'what we wonder' sticky notes — questions that will be revisited and answered across the full unit on The Cell.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Multiple lens systems] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners enter to find two sukuma wiki leaves displayed at the front: one fresh and one wilted and dried (prepared by the teacher the previous day by leaving it in the Kisumu midday sun). Working individually for 3 minutes, learners write silent responses in their science notebooks to the prompt: 'Look at these two leaves. What do you notice? What do you wonder? What do you think is happening inside the dried leaf that is not happening inside the fresh one?' Learners then share with a neighbour (Think-Pair-Share, 2 minutes) before three pairs are cold-called to share with the class. The teacher records key ideas on the board under two columns: NOTICE and WONDER, without affirming or correcting — all ideas are treated as hypotheses.	1. Before learners arrive, place the two leaves prominently at the front with a label: 'Same plant. Same garden. Why is one alive and one dead?' 2. Display the written prompt on the board and say: 'Write your thoughts in silence for 3 minutes — no talking yet. I want YOUR ideas before you hear anyone else's.' 3. After 3 minutes say: 'Now turn to your neighbour and share. You have 2 minutes.' 4. After pair-share, say: 'I am going to cold-call three pairs. Pair one — [name a learner] — what did your pair notice or wonder?' Allow 10–15 seconds WAIT TIME after each cold-call before prompting. Cold-call exactly 3 pairs. 5. Record every contribution on the board verbatim. Resist the urge to evaluate. Say: 'I am writing all ideas — right or not-yet-right — because science starts with honest thinking.' 6. Pose the bridging question to the whole class: 'If we could shrink ourselves small enough to go inside this sukuma wiki leaf, what do you predict we would see — and what would be different between the fresh leaf and the dried one?'	Think-Pair-Share with silent individual writing first (NGSS Science & Engineering Practice 1: Asking Questions). By requiring individual writing before discussion, the teacher surfaces each learner's prior knowledge without social influence. The public NOTICE / WONDER board makes thinking visible and establishes the phenomenon as a shared, puzzling event the class will explain together.	Observable indicator: Each learner has at least 2 written sentences in their notebook before discussion begins. The teacher circulates during silent writing and notes whether learners reference internal structures (prior knowledge) or focus only on external appearance. This informs whether Phase 2 microscope observations need more scaffolding on 'what to look for inside the cell.' Collect notebooks at the end of Phase 1 and scan 5 randomly selected entries.
Observe Phase 📹 **VIDEO:** [Multiple lens systems] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	In pairs, learners prepare a wet-mount slide of a very thin, hand-cut section of a fresh sukuma wiki leaf (the teacher demonstrates the cutting technique first). Pairs observe their slide under the school light microscope, starting at the lowest magnification (4×) and moving up to 40×. Each learner	1. Demonstrate wet-mount preparation on the visualiser: 'Watch my hands — I am cutting a very thin slice, almost transparent. If it is too thick, light cannot pass through and you will see nothing useful.' 2. Circulate as pairs prepare slides. For pairs struggling to get	Observation-to-Representation (NGSS SEP 3: Planning and Carrying Out Investigations; SEP 4: Analysing and Interpreting Data). By having learners draw before seeing the expert photomicrograph and then revise their drawings in a different colour, the teacher makes the gap	Observable indicator: Each learner's notebook contains two distinct labelled drawings — one from personal microscope observation and one from the electron micrograph. The teacher checks that at least 3 structures are labelled in the EM drawing (using any descriptive language).

📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	independently draws and labels what they see in a ruled circle (diameter 6 cm) in their notebook, using pencil. Labels must only name what they can actually see — they are not yet expected to use technical vocabulary. After 10 minutes of individual observation and drawing, the teacher projects a high-quality colour photomicrograph (light microscope, 400×) of a sukuma wiki leaf cell on the board. The teacher then dramatically reveals an electron micrograph (TEM, ~10 000×) of the same cell and gives learners 4 minutes to add a second drawing in their notebooks, again labelling only what they can observe.	thin sections, offer: 'Try peeling — grip the underside of the leaf and pull gently to get the lower epidermis.' 3. After pairs have observed for 5 minutes, pause the class and ask: 'Raise your hand if you can see small box-like shapes.' [Count raised hands aloud.] 'Raise your hand if you can see green dots or granules inside those boxes.' [Count again.] Say: 'Those green granules are real — they are not dirt on the lens. Remember that.' 4. Project the light micrograph. Say: 'This is what an expert prepared. Compare it with your drawing. What is the same? What did you miss? Add anything new to your drawing now — but use a different colour pencil so we can see what you added.' Give 3 minutes. 5. Now reveal the electron micrograph. Pause for 15 seconds of silent looking. Then say: 'In silence, draw what you see in a new circle. Label every structure you can see, even if you do not know its name — give it a descriptive name, like 'folded ribbons' or 'dark oval blob'.' Give 4 minutes. 6. Cold-call 2 learners to describe one structure each from the electron micrograph. Allow 10–15 seconds WAIT TIME after each question. Ask: 'What do you think that structure might be doing to keep the leaf alive?'	between prior knowledge and new evidence visible. The two-image reveal (light → electron) physically enacts the phenomenon's puzzle: the same cell looks completely different at different resolutions. Descriptive naming of unknown structures builds observational language without premature closure.	Pairs who cannot produce a usable slide are given a prepared slide — the teacher notes which pairs struggled for future differentiation. During cold-calling, listen for whether learners spontaneously link the green granules to the leaf being green or to food-making — this reveals readiness for chloroplast instruction in Lesson 3.
Explain Phase 📹 **VIDEO:** [Multiple lens systems] Source: Khan Academy (English - US curriculum)	Because this is the opening lesson, the Explain phase functions as a structured prediction activity, not a full explanation — learners do not yet have the vocabulary or knowledge to	1. Distribute or instruct learners to draw the Prediction Frame in their notebooks. Say: 'You have 5 minutes to complete at least three rows. Use the structures from your EM drawing. Make your best	Prediction Frames with deferred confirmation (NGSS SEP 6: Constructing Explanations). By structuring predictions in written form before instruction, the teacher creates cognitive hooks —	Observable indicator: Each learner has completed at least 3 rows of the Prediction Frame. The teacher collects 6 notebooks (2 high, 2 middle, 2 lower prior-knowledge learners based on Phase 1 writing)

🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	explain. Each learner independently completes a 'Prediction Frame' in their notebook (a printed template or hand-drawn table with three columns): STRUCTURE SEEN IN EM | DESCRIPTIVE NAME I GAVE IT | MY PREDICTION OF WHAT IT DOES TO KEEP THE SUKUMA WIKI ALIVE. Learners fill in at least three rows. They then participate in a brief whole-class share-out where the teacher facilitates: 'Who predicted a job for the folded membranes? Who predicted a job for the dark oval body? Who predicted a job for the large fluid space?' The teacher records class predictions on a 'Prediction Parking Lot' poster (large paper on the wall) which will remain displayed for the entire unit and be revisited each lesson. The teacher closes by connecting back to the phenomenon: 'When the sukuma wiki leaf was left in the sun and died — some of these structures stopped working. By the end of this unit, you will be able to point to each structure on this micrograph and explain exactly what it stopped doing, and why that killed the leaf — and why that matters to you, the person who eats it.'	scientific guess for the function column — there are no wrong answers right now, only guesses we will test.' 2. After 5 minutes, facilitate the share-out by pointing to specific structures on the projected EM image. 'Who wants to share their prediction for this structure — the one I am pointing to, which looks like folded ribbons?' Allow 10 seconds WAIT TIME. Cold-call if no volunteer after 10 seconds. 3. For each prediction shared, record it on the Prediction Parking Lot without evaluation. Say: 'We are parking these predictions. We will come back to check each one as we learn more.' 4. Make the unit roadmap explicit: 'In our next lessons we will identify the name and job of every structure on this image. We will answer the question: what makes up an organism — all the way from these tiny parts inside one cell, to the sukuma wiki plant in the field, to you sitting here in Kisumu.' 5. Display the unit Driving Question visually on the wall next to the Prediction Parking Lot.	learners will naturally verify their own predictions as new content is taught. The Prediction Parking Lot makes predictions semi-public and creates accountability across the unit. Connecting the phenomenon (wilting/dying leaf) to the learner as the person who eats sukuma wiki grounds cell biology in personal relevance.	and reads the predictions before the next lesson. Look for: (a) any learners who already use correct organelle vocabulary (chloroplast, mitochondria, vacuole) — these learners can serve as peer resources; (b) common misconceptions (e.g. 'the green granules are food stored in the leaf' vs 'the green granules make food') — address in Lesson 2 opener.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	The teacher introduces the Driving Question Board — a large piece of paper or section of the classroom wall that will be used throughout the entire unit on The Cell. The class reads the unit Driving Question aloud together: 'What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?'	1. Unveil the blank DQB with ceremony: 'This wall is going to be our science memory for the next several lessons. Every question we answer, we will mark. Every idea we change, we will update. It is a living record of our thinking.' 2. Read the Driving Question aloud and have learners read it back as a class. 3. Distribute paper/sticky notes.	Driving Question Board as a Knowledge-Building Community Tool (NGSS SEP 1: Asking Questions; SEP 8: Obtaining, Evaluating, and Communicating Information). The DQB externalises the class's collective prior knowledge and curiosity, making learning social and cumulative. Clustering questions shows learners that their individual wonderings are shared, building	Observable indicator: Every learner has posted at least one green and one blue card — 100% participation is required; the teacher notices any learner who has not posted and gently prompts them privately. The teacher photographs and reads all blue (WONDER) cards before the next lesson. Tally the five most common questions — these directly inform the sequencing and

📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or two small pieces of paper if sticky notes are unavailable). On the first (WHAT WE THINK WE KNOW — written in green), learners write one thing they are fairly confident about regarding cells or cell structure. On the second (WHAT WE WONDER — written in blue), learners write one genuine question they have after today's observations. Learners come forward in groups of 5 to post their notes, briefly reading each aloud. The teacher clusters similar questions spatially on the board and draws circles around clusters. The teacher names each cluster aloud ('Several of you are wondering about the green granules — great. Several are wondering about the difference between plant and animal cells — excellent.'). The DQB remains on the wall for the entire unit; the teacher commits aloud: 'Every question on this board will be answered by the time we finish Sub-Strand 1.2.'	Say: 'Take 3 minutes. Green paper: what do you already believe to be true about cells? Blue paper: what do you genuinely not understand yet after today?' Enforce silence for 3 minutes. 4. Call groups of 5 to post. As each learner reads their card, say: 'Thank you — that is an important idea/question.' Cluster on the board in real time. 5. After all cards are posted, step back and say: 'Look at this board. This is the map of our learning journey. These are the questions science has already answered — we are going to learn those answers. These are questions some scientists are still working on — exciting, right?' 6. Photograph the DQB with the school's tablet or camera for the ARES platform record. 7. Close by returning to the phenomenon: 'Every question on this board connects back to those two leaves at the front of the room. The fresh one and the dead one. By Lesson 6, you will be able to explain exactly what happened inside that dying leaf — at the level of molecules and organelles.'	intellectual community. The public commitment ('every question will be answered') creates lesson-to-lesson narrative coherence.	emphasis of Lessons 2–6. Flag any scientifically sophisticated questions (e.g. referencing DNA, proteins, or energy) for enrichment pathways.
Model Building Phase 📹 **VIDEO:** [Multiple lens systems] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)]	In the final segment, each learner draws their first 'Anchor Model' — an initial, honest representation of what they currently believe a sukuma wiki leaf cell looks like from the inside, based only on today's observations (NOT what they think they should know). The model must: (a) be drawn in pencil so it can be revised; (b) include at least 4 labelled structures using either correct names or descriptive names; (c) include a 'confidence key' — next to each label, learners	1. Distribute A3 paper or direct learners to a dedicated section of their notebooks. Say: 'This is Model Version 1. Date it. It is evidence of your thinking TODAY — before you have been taught anything. I will never grade this model for being wrong; I will grade it for being honest and detailed.' 2. After 5 minutes of drawing, circulate and look for: learners who draw only an outline box (prompt: 'What did you see INSIDE the box on the electron micrograph? Add	Anchor Model with Confidence Coding (NGSS SEP 2: Developing and Using Models). Beginning with an explicit, revisable model establishes model-based reasoning as a core practice for the unit. The confidence key (H/M/L) teaches metacognition — learners distinguish between what they know and what they are guessing, a foundational scientific habit. Referencing Dr Githeko grounds model revision in real Kenyan scientific practice,	Observable indicator: Every learner's model contains at least 4 labelled structures and a confidence key is attempted. The teacher collects Cell Model Journals at the end of the lesson and checks: (a) that internal structures (not just the cell outline) are drawn; (b) that labels reflect today's observation language (not only rote-memorised terms). Note any learner who draws an animal cell instead of a plant cell — address at the start of Lesson 2 to

Source: CK-12 🔍 Search ARES for similar readings	write H (high confidence), M (medium), or L (low). The teacher explains: 'This is Model Version 1. You will update this model every lesson. By Lesson 6 your model will be much richer and more accurate. Scientists do this — they start with an initial model and revise it as evidence grows. Dr Mwangi Githeko, the Kenyan malaria researcher at KEMRI in Kisumu, started with simple models of how the malaria parasite enters a cell — and revised his models many times as he gathered more evidence.' Learners store the Anchor Model on the first page of a dedicated 'Cell Model Journal' (a folded A3 sheet or the back pages of their science notebook).	at least 3 internal structures.'); learners using exclusively external knowledge from primary school (prompt: 'What did YOU see today under our microscope? Start with that.'). 3. After 7 minutes, cold-call 2 learners to briefly show and describe their model under the visualiser (30 seconds each). Ask the class: 'What did this learner include that you did not? Add it to your model if you agree with it.' 4. Introduce the revision protocol: 'Every time we learn something new that changes what you drew, update your model in a different colour and write the lesson number next to the change. By the end of the unit, your model should look very different from today — and that change IS the learning.' 5. Close the lesson by pointing to the two leaves: 'We started today with a simple question — why does one leaf live and one die? We now know that the answer is inside this electron micrograph. Our job across this unit is to decode it. See you next lesson.'	reinforcing that science is a process of iterative refinement, not a fixed body of facts.	ensure the phenomenon (sukuma wiki = plant cell) is properly anchored. Return journals at the start of Lesson 2.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal before the next lesson:

1. PHENOMENON UPTAKE: Did learners treat the two sukuma wiki leaves as genuinely puzzling, or did they seem disengaged from the phenomenon? If disengaged, what might make the phenomenon more personally relevant for learners in this specific school context in Kisumu? (Consider: Could you ask a learner whose family grows sukuma wiki to describe what happens to leaves in the garden?)

2. PRIOR KNOWLEDGE RANGE: What did the DQB blue cards reveal about the range of prior knowledge in the class? Are there learners who already know organelle names? Are there learners who do not yet know what a cell is? How will you differentiate Lesson 2 to address both ends?

3. MICROSCOPE READINESS: How many pairs successfully prepared a usable wet-mount slide? If fewer than 70% succeeded, plan a 5-minute slide-preparation refresher at the start of Lesson 2 before moving to new content.

4. QUALITY OF PREDICTIONS: Did learners' Prediction Frame entries show functional thinking ('the oval body produces energy') or purely structural thinking ('the oval body is brown')? Lessons 4 and 5 on organelle function should be planned with this baseline in mind.

5. DQB RICHNESS: Were the WONDER questions mostly surface-level ('what is the name of the green granule?') or conceptually deep ('if the cell has a membrane,

how does food get in?')? Deep questions signal readiness to engage with membrane transport in future units.

6. EQUITY: Did girls and boys participate equally in cold-calling and DQB posting? Note any gender patterns and use structured turn-taking in Lesson 2 if participation was uneven.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a fresh sukuma wiki leaf under the school light microscope and recorded annotated drawings of box-like compartments containing green granules. They then observed a projected transmission electron micrograph of the same leaf cell, identifying additional structures including folded membranes, dark oval bodies, and a large fluid-filled space — structures invisible under the light microscope.
What did I learn?	Learners learned that a single sukuma wiki leaf cell contains multiple distinct internal structures visible at different levels of magnification; that light microscopes and electron microscopes reveal very different levels of cellular detail; and that the gap between the two images represents unexplained complexity that motivates the unit's central question.
How does this explain the phenomenon?	Learners cannot yet fully explain what the observed structures are or what they do — and this is intentional. They have made initial predictions (recorded in their Prediction Frames and Anchor Models) connecting specific structures to possible life-sustaining functions, and they have posted genuine questions on the Driving Question Board that will be systematically answered across Lessons 2–6.

LESSON 2 (40 minutes): Light vs. Electron Microscopes — Revealing Hidden Structures in the Sukuma Wiki Cell	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will explain the key differences between light and electron microscopes — in terms of resolution, magnification, specimen preparation, and cost — and connect those differences to why the two instruments revealed such strikingly different internal detail in the same sukuma wiki cell.
Knowledge	Learners know that: (a) a light microscope uses visible light and glass lenses and can magnify up to ~1,500×; (b) an electron microscope uses a beam of electrons and electromagnetic lenses and can magnify up to ~1,000,000×; (c) resolution is the ability to distinguish two closely spaced points as separate objects; (d) electron microscopes have far superior resolution (0.1–0.2 nm vs ~200 nm for light microscopes); (e) electron microscopy requires elaborate specimen preparation (fixation, sectioning, heavy-metal staining, vacuum chamber) and is far more expensive than light microscopy; (f) light microscopes can view living or fresh specimens (e.g., a fresh sukuma wiki leaf slice) whereas electron microscopes cannot.
Skills	Obtaining and Communicating Information (SEP): learners locate relevant information from print/digital sources, extract key data, and organise it into a structured comparison table. Learners also practise scientific communication by articulating one key difference clearly and precisely to the class.
Attitudes	Learners appreciate that scientific tools vary in capability and that choosing the right tool for the right question is a scientific skill. Learners respect peers' contributions when cold-called and remain intellectually curious about what additional structures the electron microscope reveals inside the sukuma wiki cell.
Key Inquiry Question	What makes up an organism?
Purpose in Storyline	In Lesson 1, learners anchored their curiosity in the phenomenon — a fresh sukuma wiki slice showing green-granule-filled compartments under the school light microscope, versus far richer internal detail in the electron photomicrograph. Lesson 2 is the first evidence-gathering lesson: before learners can explain what all those internal structures do, they must understand WHY the two instruments show such different pictures. By the end of this lesson, learners can resolve the first layer of the puzzle (different instruments → different resolution → different detail) and are primed to ask: so what ARE all those extra structures the electron microscope reveals?
Safety Notes	No hazardous chemicals or electrical equipment are used in this lesson. Learners handle print materials and/or school digital devices. If devices are used, remind learners to follow the school's cyber-safety policy and not to open unverified external links. Learners with visual impairments should be paired with a sighted partner and given large-print comparison tables.

B. LESSON OVERVIEW

Lesson 2 is the first full Observe-phase lesson in the 8-lesson sequence. Working in pairs, learners use print media (teacher-prepared fact-sheets) and/or school digital devices to research the differences between light and electron microscopes across four dimensions: resolution, magnification, specimen preparation, and cost/accessibility. Each pair records findings in a structured two-column comparison table. The teacher then cold-calls four pairs to share one key difference each, building a class master table on the board. The lesson closes by explicitly reconnecting each microscope difference to the phenomenon: why did the Kisumu learner see only green granules in compartments under the light microscope, yet folded membranes, dark oval bodies, and a large fluid-filled space under the electron microscope? This lesson addresses KICD SLO (a) — explaining the differences between a light and an electron microscope — and embeds the NGSS Science and Engineering Practice of Obtaining and Communicating Information.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation: Light Years to the Centaurus Constellation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two side-by-side images on a printed A3 sheet (or projected on screen): Image A — the school's own light-microscope photomicrograph of a fresh sukuma wiki leaf cross-section showing green-granule-filled box-like compartments; Image B — an electron-microscope photomicrograph of the same leaf type showing folded internal membranes, dark oval bodies, and a large fluid-filled space. Learners individually write responses (2 minutes, silent) to the prompt: 'These two images are of the SAME type of cell. List THREE differences you notice between them. Why do you think the two pictures look so different — what might be different about the two instruments used?' Learners do not share yet.	Display both images prominently. Read the prompt aloud slowly. Say: 'Do not discuss with your partner yet — this is YOUR thinking first. You have 2 minutes. Write at least three observations and one idea about why the images look different.' Set a visible timer. After 2 minutes, say: 'Hold your predictions — we will revisit them at the end of the lesson to see how your thinking has changed.' Do NOT correct or validate predictions at this stage.	Activate Prior Knowledge + Prediction Journaling. By committing initial ideas to paper before instruction, learners create a cognitive anchor. When new information arrives during the Observe phase, they can explicitly compare it to their prediction, deepening encoding and metacognitive awareness.	Circulate and scan written predictions (do not collect). Look for: (i) whether learners notice resolution/clarity differences rather than only colour differences; (ii) whether any learner spontaneously uses the word 'magnification' or 'resolution'. This gives the teacher real-time data on prior knowledge to calibrate the depth of explanation needed in the Explain phase.
Observe Phase 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation: Light Years to the Centaurus Constellation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs. Each pair receives a teacher-prepared one-page Fact-Sheet containing: (a) a labelled diagram of a light microscope with key specifications (max magnification ~1,500×, resolution ~200 nm, uses visible light, can view living specimens, costs KES 5,000–50,000 in Kenyan school markets); (b) a labelled diagram of a transmission electron microscope with key specifications (max magnification ~1,000,000×, resolution ~0.1–0.2 nm, uses electron beam, requires vacuum and dead/fixed specimens, costs upwards of KES 50,000,000); (c) two annotated photomicrographs — the same sukuma wiki leaf under each	Distribute fact-sheets and blank comparison tables simultaneously. Say: 'In pairs, use the fact-sheet — and a device if you have one — to fill in every row of your comparison table. You have 12 minutes. Both partners must be able to explain any row — I will cold-call individuals, not just pairs.' Circulate every 2–3 minutes. Probe pairs that are stuck on resolution with: 'Resolution is not how big the image looks — it is how much DETAIL you can see. If two points are 0.5 nm apart, which microscope can tell them apart as two separate points?' [Allow 10-second WAIT TIME after this question before offering a hint.] If a pair finishes early, prompt: 'Look at	Structured Paired Inquiry with a Comparison Table (Obtaining and Communicating Information — SEP). The comparison table scaffolds parallel processing: learners must hold information about BOTH instruments in the same cognitive frame, making contrasts salient rather than treating each microscope in isolation. The table format also prepares learners for the cold-call share-out, reducing anxiety by giving them a written reference.	Check that each pair's table has a correct, specific value in the Resolution row (not vague phrases like 'very clear' but actual nm values). If more than 3 pairs are missing numerical resolution values after 8 minutes, pause the class and say: 'Freeze — everyone look at row 3. Resolution is measured in nanometres. Find the number on your fact-sheet and write it now.' This is the pivotal scientific concept of the lesson and must be correct before the class share-out.

	instrument. If school devices are available, pairs may also search 'light vs electron microscope comparison' on the Kenya Education Cloud or approved biology resource sites. Each pair fills in a two-column comparison table with rows: (1) Energy/Beam Used, (2) Maximum Magnification, (3) Resolution, (4) Specimen Condition (living/dead), (5) Specimen Preparation Steps, (6) Approximate Cost in Kenya, (7) One Advantage, (8) One Limitation. They have 12 minutes.	row 4 — Specimen Condition. What does that mean for the sukuma wiki leaf in the phenomenon? Could the electron microscope have been used on the SAME fresh leaf the learner was looking at?'		
Explain Phase 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation: Light Years to the Centaurus Constellation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Teacher-facilitated class share-out and sensemaking discussion (10 minutes). The teacher draws a blank two-column table on the board (Light Microscope | Electron Microscope) and cold-calls FOUR pairs — each pair shares ONE row they found most interesting or surprising. After all four contributions are on the board, the teacher leads a whole-class discussion linking each difference back to the phenomenon: 'We now have four key differences. Let us connect each one back to our sukuma wiki leaf in Kisumu.' Learners then individually write a 2-sentence explanation in their exercise books answering: 'Why did the electron microscope reveal folded membranes, dark oval bodies, and a large fluid-filled space that were invisible under the school light microscope?'	Cold-call sequence (name four pairs before class — ensure gender balance and mix of confidence levels): 'Pair 1 — [names] — tell us one row from your table. Start with the row you found most surprising.' After the pair responds, allow 10 seconds of WAIT TIME, then ask the class: 'Does anyone want to add to or refine what [names] said?' Repeat for Pairs 2, 3, and 4. After all four contributions, point to the phenomenon images and ask: 'Given what is now on our board — which single difference explains MOST clearly why the two microscope images of the sukuma wiki cell look so different?' [Allow 15 seconds WAIT TIME — this is a higher-order question.] Anticipate the answer 'resolution' and affirm: 'Exactly — the electron microscope's resolution of about 0.1 nm means it can resolve structures that are nanometres apart, while the light microscope cannot resolve anything closer than about 200 nm. That is why the folded membranes — which are only a few nanometres thick — are invisible under the light	Cold-Call Jigsaw Share-Out + Phenomenon Re-Anchoring. Each cold-called pair contributes one unique row, so the class co-constructs the full master table collaboratively — no one pair holds all the information. Re-anchoring each difference to the sukuma wiki phenomenon prevents the comparison table from becoming an abstract fact-list and keeps learning tethered to the driving question.	Collect the two written sentences from 6–8 learners (a random sample) as exit-ticket evidence. A proficient response must: (a) mention resolution (not just magnification) and (b) explicitly state that the light microscope cannot resolve the nanometre-scale internal structures. Flag any response that only says 'the electron microscope is more powerful' as insufficient — these learners need a follow-up probe in Lesson 3.

		microscope.' Then say: 'Now write two sentences in your book explaining this to the Kisumu learner in the phenomenon.' Give 3 minutes of silent individual writing time.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation: Light Years to the Centaurus Constellation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each pair writes on a sticky note (or a torn piece of paper): one NEW question that today's lesson has raised for them about the sukuma wiki cell — something they now want to know about the structures visible in the electron microscope image. Examples might include: 'What are the folded membranes doing?', 'Why are the oval bodies dark in the electron image?', 'Does the large fluid space do anything important?' Pairs post their sticky notes on the class DQB under the column 'What we still need to find out'. The teacher reads 3–4 notes aloud and says: 'These questions about the internal structures — the folded membranes, the oval bodies, the fluid-filled space — are exactly where we are going next. Lessons 3 onwards will help us name and explain every one of these structures.'	Distribute sticky notes. Say: 'Based on what you now know about WHY the electron microscope shows more detail, write one new question you have about what you can actually SEE in the electron microscope image — one of those structures you want to understand. You have 2 minutes.' After posting, read 3–4 notes aloud without attribution. Say: 'Notice that many of our new questions are about the STRUCTURES inside the cell — the folded membranes, the dark oval shapes, the fluid space. Those are cell organelles. The next lessons will let us answer these questions one by one. Keep these questions in mind as we go forward.' Do not answer the organelle questions yet.	Driving Question Board — Question Sorting and Progression Tracking. The DQB makes the knowledge-building arc visible to learners. By generating new questions today, learners experience that good science does not close questions but opens new, more precise ones — a key disposition of scientific thinking.	Scan the posted sticky notes for conceptual quality. A strong note asks a specific structural or functional question (e.g., 'What do the folded membranes in the sukuma wiki cell do?'). A weak note repeats today's content (e.g., 'What is an electron microscope?'). If more than half the notes are weak/repetitive, use the first 5 minutes of Lesson 3 to model what a strong DQB question looks like by referencing the phenomenon.
Model Building Phase 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation: Light Years to the	Learners update their personal running model of the sukuma wiki cell (begun in Lesson 1 or started fresh if this is the first model-building opportunity). They draw two side-by-side boxes labelled 'What the light microscope shows' and 'What the electron microscope shows'. In the light-microscope box they draw the box-like compartment with green granules only. In the electron-microscope box they draw the same compartment but add: folded membranes (labelled '?'), dark oval	Say: 'In the last 3 minutes, update your cell model. Draw the two boxes as I described. Use question marks for the structures you cannot yet explain — those question marks are your personal version of our DQB. They remind you of what you are still trying to figure out.' Circulate and check that learners are drawing both boxes (not just one). Prompt learners who draw only one box: 'Show me BOTH instruments' view of the same cell — that contrast is the point of today's lesson.' Close	Cumulative Model Revision — Annotated Dual-View Drawing. Making the model explicitly show what each instrument can and cannot reveal forces learners to internalise that observation tools constrain and shape what counts as evidence. The question-mark labels create productive cognitive tension that motivates the next lessons.	Quickly scan models as you circulate. Look for: (a) two distinct boxes (not one); (b) the electron-microscope box containing at least three additional structures (folded membranes, oval bodies, fluid space); (c) question-mark labels on the new structures. Absence of (b) or (c) indicates the learner has not yet connected the resolution concept to the visible structural difference — address individually at the start of Lesson 3.

Centaurus Constellation (PLIX) Source: CK-12 🔍 Search ARES for similar readings	bodies (labelled '?'), and a large fluid-filled space (labelled '?'). The question marks signal structures not yet explained — to be filled in across subsequent lessons. Learners write beneath the model: 'The electron microscope reveals more structures because its resolution (~0.1 nm) is far greater than the light microscope (~200 nm). The extra structures are too small for the light microscope to resolve.'	the lesson by saying: 'Today you solved the first mystery: why the two microscope images look so different. Tomorrow we start naming and explaining every structure with the question mark.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Cold-call quality — did the four pairs share genuinely different rows, or did pairs 3 and 4 repeat what pairs 1 and 2 had already said? If repetition occurred, restructure the share-out next time by assigning specific rows to specific pairs in advance. (2) Resolution vs. magnification confusion — did most learners' exit-ticket sentences mention resolution specifically, or did they conflate it with magnification? If magnification dominated, plan a brief 3-minute clarification at the start of Lesson 3 using the analogy: 'A Nairobi street photographer can zoom into a crowd (magnification) but if the lens is blurry, zooming in only shows a bigger blur (poor resolution).' (3) DQB quality — were the new questions specific and structural (naming the folded membranes, oval bodies, fluid space) or vague? If vague, model a strong question in Lesson 3. (4) Pacing — the lesson has five timed segments; note which ran over and adjust Lesson 3 accordingly. (5) Equity check — were girls and boys cold-called equally? Were learners from different counties or language backgrounds equally heard? Adjust cold-call list for Lesson 3 if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners studied a teacher-prepared fact-sheet and/or digital sources comparing light and electron microscopes, and completed a two-column comparison table covering beam type, maximum magnification, resolution, specimen condition, specimen preparation, approximate cost in Kenya, one advantage, and one limitation.
What did I learn?	Learners can explain that the electron microscope's far superior resolution (~0.1–0.2 nm vs ~200 nm for the light microscope) is the primary reason it reveals nanometre-scale internal structures — folded membranes, dark oval bodies, and the large fluid-filled space — in the sukuma wiki cell that are completely invisible under the school light microscope.
How does this explain the phenomenon?	Learners updated their running cell model with two side-by-side views (light vs. electron microscope) of the sukuma wiki cell, labelling the extra structures with question marks to signal unresolved explanations, and posted new specific questions on the Driving Question Board about what those structures do — questions that will drive evidence-gathering in Lessons 3–8.

LESSON 3 (80 minutes (2 × 40-minute lessons)): Animal Cell Ultrastructure: Observing the Hidden Machinery of Life	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to observe, identify, and explain the ultrastructure of an animal cell as seen under the electron microscope, connecting each organelle's function to the survival of living organisms.
Knowledge	Learners describe the structures of an animal cell as seen under the electron microscope — including the cell membrane, nucleus, nucleolus, nuclear envelope, mitochondria, ribosomes, rough and smooth endoplasmic reticulum, Golgi apparatus, lysosomes, centrioles, and cytoplasm — and explain the function of each component.
Skills	Learners observe and annotate electron micrograph photomicrographs of animal cells (human cheek cell); draw and label accurate diagrams from photomicrographs; compare visible ultrastructure details with what was seen under the light microscope in Lesson 2; and record evidence in a structured observation table.
Attitudes	Learners appreciate the cell as a highly organised, functional unit of life and show respect for peers' annotations and interpretations during group work, recognising that every organelle contributes to keeping an organism — from the sukuma wiki plant to the human learner — alive.
Key Inquiry Question	What makes up an organism?
Purpose in Storyline	In Lesson 2, learners observed the plant cell under the electron microscope and identified organelles unique to the plant cell. In Lesson 3, learners now shift focus to the animal cell — specifically a human cheek cell — observed under the electron microscope. This lesson advances the driving question by giving learners the second half of the ultrastructural evidence they need: what organelles does an animal cell possess, what are their functions, and how do these compare with the plant cell? By the end of this lesson, learners will have enough evidence to begin explaining how the tiny structures inside a single cell keep a whole organism — including themselves as human beings in Kisumu or Nairobi — alive. The DQB is updated to reflect this new evidence layer.
Safety Notes	No hazardous chemicals or sharp instruments are used in this lesson. Learners handle printed or digital photomicrograph images only. If digital devices are shared, learners must handle them with clean, dry hands. No food or drink near any devices. The teacher should ensure equitable access to photomicrographs across all groups and that no learner is excluded from the annotation activity.

B. LESSON OVERVIEW

Learners work in groups of three to observe and annotate high-resolution electron micrograph photomicrographs of a human cheek cell, identifying organelles visible at the ultrastructural level. Through a structured Observe–Record–Connect sequence, each group produces a labelled diagram. The teacher then facilitates a whole-class discussion using Kenyan analogies to anchor each organelle's function (mitochondria = jiko/charcoal stove; nucleus = county governor's office; ribosomes = village artisans; Golgi apparatus = Nairobi GPO post office; lysosomes = municipal waste-disposal trucks). Learners compare the animal cell ultrastructure with the plant cell from Lesson 2, identifying similarities and differences. The lesson concludes with an update to the Driving Question Board and a brief formative exit task. This lesson addresses KICD SLO (b): describe the animal cell under an electron microscope, and advances SLO (c): compare plant and animal cell structures.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually study the same sukuma wiki plant cell electron micrograph from Lesson 2 (displayed on the board or in their ARES workbook). They then look at a deliberately low-resolution, partially blurred outline of a human cheek cell electron micrograph. Without any teacher input, each learner writes answers to two predict prompts in their science journal: (1) 'List THREE structures you expect to see inside a human cheek cell that you also saw in the sukuma wiki plant cell.' (2) 'List TWO structures you expect will be MISSING from the human cheek cell that were present in the plant cell. Give a reason for each.' Learners share predictions with their group partner (Think-Pair-Share, 2 minutes) before the teacher takes a whole-class show-of-hands poll on each prediction.	Teacher displays the blurred cheek-cell outline on the board alongside the plant cell EM from Lesson 2 and says: 'Last lesson we looked inside a sukuma wiki plant cell using the electron microscope. Today we are looking at a cell from inside your own body — a human cheek cell. Before we look at the real image, I want you to use the evidence you already have.' Teacher reads both prompts aloud and gives 4 minutes of SILENT individual writing time (no discussion). Teacher circulates, reading over shoulders without commenting. After 4 minutes, teacher says: 'Now share your two predictions with the person next to you — 2 minutes.' After sharing, teacher conducts a cold-call poll: 'Hands up — who predicted they would see a nucleus?' (tally on board). Teacher cold-calls 3 learners who did NOT raise their hand: 'Achieng, you didn't raise your hand — what did you predict instead, and why?' WAIT TIME: 12 seconds before prompting. Teacher records class predictions in two columns on the board ('Structures we expect to FIND' / 'Structures we expect to be ABSENT') and says: 'We are going to test every one of these predictions against the actual electron micrograph. Keep your journals open — you will need to revise these predictions.'	Think-Pair-Share with whole-class prediction tally. This strategy activates prior knowledge from Lesson 2 (plant cell ultrastructure) and creates cognitive anticipation — learners are invested in checking whether their predictions are confirmed or refuted by the real EM image. Recording predictions publicly on the board creates accountability and sets up the 'prediction vs. evidence' tension that drives the Observe phase.	Teacher scans written predictions during circulation (4-minute silent period). Observable indicator: each learner has written at least 3 expected structures and 2 expected absent structures with a reason. Teacher notes learners who leave the 'reason' column blank — these learners will be targeted with a direct question during Phase 2. The board tally reveals class-wide prior knowledge gaps (e.g., if fewer than 50% predict absence of chloroplasts, teacher knows this misconception must be addressed explicitly in the Explain phase).
Observe Phase 📹 **VIDEO:** Multiple lens systems Source: Khan	Each group of three receives: (a) a printed A4 high-resolution electron micrograph of a human cheek cell with a scale bar, and (b) a blank A5 'Observation Record Table' with columns: Organelle Name |	Teacher distributes photomicrographs and record tables and says: 'You have 15 minutes as a group. Your only job right now is to identify what you can SEE — do not worry about	Structured Observation with Role Assignment and an Identification Key scaffold. Assigning roles (Observer, Recorder, Checker) ensures all group members engage with the photomicrograph	Teacher checks observation record tables during circulation. Observable indicators: (1) table has at least 6 organelles identified with a descriptive EM appearance phrase; (2) individual diagram has

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Sketch of appearance in EM | Location in cell | Inferred function (leave blank for now). Groups work through the photomicrograph systematically, using a printed 'EM Organelle Identification Key' (provided on ARES platform — a simplified visual guide showing characteristic EM appearances of 9 organelles). Each group member takes a role: Observer (points to structures on the image), Recorder (fills in the table), and Checker (cross-references the Identification Key). After 15 minutes of group observation, each learner independently draws a large labelled diagram of the cheek cell in their science journal, adding as many organelles as the group identified. Learners annotate with the organelle name and one descriptive phrase about its EM appearance (e.g., 'mitochondria — oval, double membrane, folded inner membrane called cristae').	function yet, that comes later. Use the Identification Key. Every structure you circle on the image must go into your table.' Teacher sets a visible 15-minute timer. Teacher circulates to all groups at least twice, spending no more than 90 seconds per group. At each group, teacher asks ONE targeted question from this list (rotating): 'What is the double-layered boundary around the entire cell called?', 'Can you see any folded membrane structures inside the large oval body near the centre?', 'How many mitochondria can your group count in this image?', 'What is different about the membrane around the nucleus compared to the cell membrane?' Teacher does NOT give answers — responds to correct identifications with 'Good — record that' and to incorrect identifications with 'Look at the Key again — compare the shape described there with what you are seeing.' At the 15-minute mark, teacher calls time: 'Pens down on the group table. Now, individually, every person draws and labels their own diagram — 8 minutes, silent.' During individual drawing, teacher cold-calls 2 learners to share one organelle identification with the class: 'Otieno, tell us one structure your group found and describe what it looks like in the image.' WAIT TIME: 10 seconds. Teacher records each shared organelle on the board in a running class list.	rather than one learner dominating. The Identification Key provides just enough scaffold to make the task achievable without pre-teaching — learners must still match visual features themselves. The transition from group table to individual diagram forces each learner to personally process and represent the evidence, deepening encoding.	at least 6 labelled structures with legible labels and leader lines that point to the correct location. Teacher flags any group that has fewer than 4 organelles after 10 minutes and provides a prompt: 'Look at the large dark oval shape near the centre of the image — what does the Key say about that shape?'
Explain Phase 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English	Teacher leads a whole-class 'Organelle–Job–Analogy' discussion. For each of 9 organelles, learners first propose a function based on its structure (using their observation table),	Teacher says: 'Now we are going to find out what each of these structures is actually doing — because knowing the name means nothing if you do not know the job.' Teacher works through organelles	Analogy-Based Sensemaking (Structure–Function–Analogy triads). By requiring learners to first propose a function before hearing it, the teacher surfaces misconceptions and ensures	Teacher listens to cold-called responses during the function-proposal step. Observable indicator: learner can link at least one structural feature observed in the EM (e.g., 'it has a double

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	then the teacher confirms/corrects and introduces a Kenyan analogy. Learners record the confirmed function and analogy in a third column of their observation table. After the teacher-guided discussion (15 minutes), learners individually complete a 'Comparison Table' in their journals with three columns: Organelle | Present in Plant Cell (✓/✗) | Present in Animal Cell (✓/✗). This directly revisits their Lesson 2 plant cell observations and begins to build SLO (c). Learners then answer one written explanation prompt: 'A human being in Nairobi eats ugali and nyama choma. Explain how the mitochondria in the cheek cells of that person are connected to the energy released from that meal.' Groups share responses (2 minutes peer discussion) before one response is shared with the class.	one by one on the board, using the following sequence for each: (1) Teacher points to the organelle on the displayed EM and asks: 'Based on its structure — what do you think this might be doing? 10 seconds to think, then I will cold-call.' WAIT TIME: 10 seconds. Cold-calls 2 learners per organelle (targeting learners identified as needing support in Phase 1). (2) Teacher confirms or corrects the function with a brief scientific explanation. (3) Teacher introduces the Kenyan analogy and writes it on the board. Analogies used: Nucleus = 'County Governor's office in Kisumu — it holds all the instructions for running the cell, just as the governor's office holds the plans for running the county'; Mitochondria = 'The jiko — the charcoal stove in your kitchen at home. It takes the ugali and nyama choma you ate and releases the energy stored in that food'; Ribosomes = 'Village artisans (fundis) — tiny but essential, they build the proteins the cell needs from raw materials'; Rough ER = 'The road network connecting the fundi's workshop to the delivery point — proteins made on ribosomes travel along it'; Golgi Apparatus = 'The Nairobi GPO (General Post Office) — it packages and addresses proteins and sends them to the right destination inside or outside the cell'; Lysosomes = 'Nairobi City Council waste-disposal trucks — they break down and recycle old, damaged cell parts'; Cell membrane = 'A security gate at a Mombasa port — it controls what enters and leaves the cell'; Cytoplasm = 'The busaa/porridge	learners construct meaning rather than passively receive it. The Kenyan analogies (jiko, county governor, GPO, City Council trucks) ground abstract ultrastructural concepts in familiar local contexts, making them memorable and culturally relevant. The comparison table enforces cross-lesson synthesis, linking Lesson 2 and Lesson 3 evidence.	membrane with folds') to a proposed function (e.g., 'maybe it is for making energy'). Written ugali/mitochondria prompt is collected by teacher at the end of the phase as a written formative check. Success criterion: response must mention (a) food as an energy source, (b) mitochondria as the site of energy release, and (c) a connection to the cell's need for energy to carry out its functions.

		base — everything floats and operates within it'; Nucleolus = 'The records room inside the governor's office — it produces the instructions (rRNA) for making ribosomes.' After the 15-minute discussion, teacher gives the comparison table task and the ugali/mitochondria prompt, sets 8 minutes, and circulates. Teacher cold-calls one learner to read their ugali/mitochondria response and asks the class: 'Does anyone want to add to or challenge what Wanjiku just said?' WAIT TIME: 10 seconds.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Multiple lens systems] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky-note contributions for the class DQB on the question: 'What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?' Sticky note 1 (GREEN): 'Something I can now explain that I could not explain before this lesson.' Sticky note 2 (ORANGE): 'A new question this lesson has raised for me.' Learners physically place their sticky notes on the DQB under the appropriate organiser columns: 'Evidence we have gathered' | 'Connections we have made' | 'Questions still unanswered'. The teacher reads 3–4 orange sticky notes aloud to the class and places them prominently on the 'Questions still unanswered' column, linking them to upcoming lessons: e.g., 'Wambua asks: if the mitochondria release energy from food, how does the food get broken down first? — that is exactly what we will investigate in Sub-Strand 1.3, Nutrition in Animals.'	Teacher distributes green and orange sticky notes and says: 'The DQB is our running record of what we are figuring out together. Two contributions each — 4 minutes, silent writing.' WAIT TIME enforced: teacher does not speak during the 4-minute writing period. After 4 minutes, teacher says: 'Place your green note under Evidence, your orange note under Questions.' Teacher circulates as learners post, reading notes. Teacher selects 3 orange notes to read aloud (choosing notes that bridge to future lessons in the sequence). Teacher updates the DQB header to show 'Lesson 3 of 8 — Animal Cell Ultrastructure COMPLETE' and draws an arrow from the animal cell evidence column to the plant cell evidence column posted in Lesson 2, saying: 'Look — we now have both sides of the comparison. Lesson 4 will put these two sides together formally.'	Public Knowledge-Building on a Driving Question Board. The DQB externalises the class's collective growing understanding and makes the cumulative nature of the evidence-gathering sequence visible. By requiring both a 'can now explain' note and a 'new question' note, the teacher ensures learners engage metacognitively — recognising both what they have learned and what remains unresolved. The teacher's act of linking orange notes to future lessons demonstrates that unanswered questions are scientifically valuable, not failures.	Teacher reads all green sticky notes after the lesson. Observable indicator for success: the green note references a specific organelle by name AND links it to a function OR to the phenomenon (e.g., 'I can now explain that the mitochondria in the cheek cells of a person in Nairobi release energy from ugali'). Green notes that only say 'I learned about cells' are flagged — those learners will be individually checked at the start of Lesson 4.

Model Building Phase 📹 **VIDEO:** Multiple lens systems Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative cell model diagram begun in Lesson 1 (a hand-drawn or ARES digital model representing their initial ideas about cell structure). Each learner now adds a second layer to their model: they draw a human cheek cell adjacent to (or overlapping with) their earlier plant cell sketch, labelling all 9 organelles identified today. Using a coloured pencil or digital annotation tool (ARES), learners mark organelles present in BOTH cells (e.g., nucleus, mitochondria, cell membrane, ribosomes) in BLUE and organelles present ONLY in the animal cell (e.g., centrioles, lysosomes) in RED. Learners add one written sentence at the bottom of their model page: 'Both the sukuma wiki plant cell and the human cheek cell contain [list 3 shared organelles], which shows that all living cells share the same basic machinery for staying alive.'	Teacher says: 'Open your cumulative model — the one you started in Lesson 1. Today you are adding the animal cell as the second layer. You have 8 minutes. Use BLUE for shared organelles, RED for animal-cell-only organelles.' Teacher circulates and makes two targeted model-revision prompts per learner group: (1) 'I can see you have drawn a cell wall on your animal cell — does the evidence from today's EM support that?' (WAIT TIME: 10 seconds for learner to self-correct.) (2) 'Your nucleus is labelled — can you now add the nucleolus inside it and the nuclear envelope around it, because we saw both of those in the EM today.' In the final 2 minutes, teacher holds up one well-annotated model (with learner's permission) and says: 'Notice how Adhiambo has drawn the double membrane of the mitochondrion — that structural detail is what makes the jiko analogy work. The folds — the cristae — are like the grate of the jiko that increases the surface for heat production.' Teacher closes by connecting back to the phenomenon: 'The sukuma wiki leaf wilts and dies in the sun. The learner in Kisumu wondered why. We now know that inside every leaf cell and inside every cell of the learner's own body, there are mitochondria releasing energy, a nucleus holding instructions, and ribosomes building proteins. When those structures stop working — the cell dies. When enough cells die — the leaf wilts. We are building the evidence to answer our driving question, piece by piece.'	Cumulative Model Revision. Requiring learners to physically annotate and colour-code a running model (rather than starting a new diagram each lesson) makes the progressive nature of scientific understanding visible. The colour-coding task (blue = shared, red = animal-only) is a low-stakes but cognitively demanding comparison exercise that directly operationalises KICD SLO (c). The teacher's closing connection to the wilting sukuma wiki phenomenon re-anchors all new knowledge to the original motivating observation.	Teacher collects or photographs cumulative models at the end of the lesson. Observable indicators: (1) at least 6 organelles labelled correctly on the animal cell diagram; (2) colour-coding is present and consistent; (3) the closing sentence correctly names at least 3 shared organelles. Any model missing centrioles or lysosomes (the two animal-cell-specific organelles covered today) is noted — the teacher will re-address these at the start of Lesson 4 during the DQB review.

D. TEACHER REFLECTION
After this lesson, reflect on the following questions in your professional journal: 1. EVIDENCE QUALITY: When learners observed the electron micrograph in Phase 2, were they identifying organelles from genuine structural reasoning, or were they pattern-matching from the Key without understanding? In your next lesson, how will you check whether learners can identify an organelle in a NEW, unfamiliar EM image without the Key? 2. ANALOGY EFFECTIVENESS: The Kenyan analogies (jiko, county governor, GPO) are designed to make function memorable. However, analogies can also create misconceptions — for example, learners might think the mitochondria literally 'burn' food. Did any learners show evidence of over-literal analogy thinking in their written ugali/mitochondria prompt? If so, how will you address this in Lesson 4? 3. EQUITY OF PARTICIPATION: Cold-calling and role assignment in groups are designed to ensure all learners engage. Review your notes: which learners were NOT cold-called today? Which learners dominated group roles? In Lesson 4 (Plant vs. Animal Cell Comparison), rotate group roles deliberately and ensure learners who were quiet today are assigned the 'spokesperson' role during the class share-out. 4. DQB ORANGE NOTES: Read all orange sticky notes before Lesson 4. Identify the 2–3 most common unanswered questions. If more than 30% of learners raised a question about HOW the nucleus 'stores instructions' (i.e., DNA/chromosomes), consider adding a 5-minute 'bridge explanation' at the start of Lesson 4, as this concept underpins the comparison activity. 5. PHENOMENON CONNECTION: Did learners make the explicit connection between organelle function and the wilting sukuma wiki phenomenon unprompted, or did the connection only appear when the teacher made it in Phase 5? If learners struggled to make this connection independently, consider adding a 'Phenomenon Check' sticky note task to Phase 4 of Lesson 4: 'Write one sentence connecting today's evidence to why the sukuma wiki leaf wilts and dies.'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Electron micrograph (EM) photomicrographs of a human cheek cell showing: cell membrane, nucleus with nuclear envelope and nucleolus, mitochondria with cristae, ribosomes, rough and smooth endoplasmic reticulum, Golgi apparatus, lysosomes, centrioles, and cytoplasm — all at a level of detail impossible to see under the light microscope used in Lesson 1.
What did I learn?	An animal cell (e.g., human cheek cell) contains at least 9 identifiable organelles under the electron microscope, each with a specific structure linked to a specific function: the nucleus (county governor's office) holds genetic instructions; mitochondria (jiko/charcoal stove) release energy from food such as ugali and nyama choma; ribosomes (village fundis) build proteins; the Golgi apparatus (Nairobi GPO) packages and dispatches materials; lysosomes (City Council waste trucks) break down waste; the cell membrane (Mombasa port security gate) controls entry and exit. Animal cells differ from plant cells in lacking a cell wall, chloroplasts, and a large central vacuole, but having centrioles and lysosomes.
How does this explain the phenomenon?	The reason the sukuma wiki leaf wilts and dies — and the reason the human learner in Kisumu stays alive — is that every cell in both organisms contains a set of working organelles that together maintain the cell's functions. When these organelles cease to function (due to lack of water, nutrients, or energy), the cell dies, then the tissue dies, then the organ dies, then the organism dies. The electron microscope reveals this hidden machinery in a level of detail the light microscope cannot — which is why the two microscopes in Lesson 1's phenomenon appeared to show 'different' cells: they were revealing different levels of the same cellular

	reality.

LESSON 4 (40 minutes): Plant Cell Ultrastructure — Observing What Makes a Sukuma Wiki Cell Uniquely a Plant Cell	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners observe electron micrograph photomicrographs of plant cells (sukuma wiki and onion) to identify structures unique to plant cells, then construct a Venn diagram comparing plant and animal cells, advancing their explanation of why the sukuma wiki leaf looks different under the two microscopes.
Knowledge	– Describe the ultrastructure of a plant cell as seen under the electron microscope, naming at least eight organelles (cell wall, cell membrane, chloroplast, mitochondrion, nucleus, rough endoplasmic reticulum, large central vacuole, tonoplast, plasmodesmata). – Identify three structures present in plant cells but absent in animal cells: cell wall, chloroplasts, and a large central vacuole. – Explain how each plant-specific structure contributes to keeping the sukuma wiki leaf alive (structural support from cell wall, photosynthesis in chloroplasts, turgor pressure from vacuole).
Skills	– Analyse and annotate electron micrograph photomicrographs of plant cells to identify and label organelles accurately. – Construct a Venn diagram comparing plant and animal cell ultrastructures using evidence from electron micrographs.
Attitudes	– Appreciate that the unique structures of plant cells explain observable phenomena — such as why a sukuma wiki leaf is green, rigid when fresh, and collapses when it dries. – Value collaborative evidence-gathering and respectful peer comparison of observations.
Key Inquiry Question	What makes up an organism — and specifically, which internal structures make a plant cell different from an animal cell, and how do those differences keep a sukuma wiki plant alive?
Purpose in Storyline	In Lesson 3 learners mapped the internal machinery of an animal cell. Lesson 4 now places the sukuma wiki cell centre-stage: learners gather direct visual evidence from electron micrographs of plant cells, identify the three structures the animal cell lacks, and begin to explain why the wilting, drying, and green colour of the sukuma wiki leaf in the anchoring phenomenon are all connected to specific organelles — bridging observations from the light microscope (green granules, box-like compartments) to the far richer ultrastructure revealed by the electron microscope.
Safety Notes	No chemical or biological hazards in this lesson. Learners handle printed or digital photomicrographs only. Remind learners to handle printed materials carefully and to use rulers or pencils — not fingers — when pointing to structures on shared prints to maintain hygiene.

B. LESSON OVERVIEW

This lesson is the pivot point in the evidence-gathering sequence: having built a detailed map of animal cell ultrastructure in Lesson 3, learners now turn their gaze to the cell at the heart of the anchoring phenomenon — the sukuma wiki plant cell. The lesson is structured as a true Observe phase in the ARES Predict–Observe–Explain–DQB–Model framework, centred on careful analysis of electron micrograph photomicrographs of plant cells (sukuma wiki leaf mesophyll and onion epidermal cells). Learners work in groups of four to annotate two photomicrographs, adding a "Plant Cell" column to the comparative diagram table begun in Lesson 3. The deliberate juxtaposition of the two photomicrographs — one a low-magnification transmission electron micrograph showing the whole cell, the other a high-magnification image focusing on a chloroplast with visible thylakoid membranes — mirrors the two-microscope contrast at the heart of the phenomenon and reinforces why the electron microscope is necessary to understand how the cell keeps the leaf alive.

The lesson builds directly on learners' prior knowledge. In Lesson 2 they established that the light microscope could reveal box-like compartments and green granules in

the sukuma wiki cell, but could not resolve internal membrane detail. In Lesson 3 they identified nine animal cell organelles and noted that animal cells lack a cell wall, chloroplasts, and a large central vacuole. Lesson 4 asks: what exactly are those three missing structures, what do they look like at the ultrastructural level, and what do they do? The Venn diagram comparison — co-constructed as a whole class at the end of the Observe phase and consolidated in the Explain phase — makes the structural differences visible, concrete, and memorable, connecting each plant-specific structure back to a visible property of the fresh sukuma wiki leaf. By the end of the lesson, learners can explain three key phenomena: (1) why the sukuma wiki leaf is green — chloroplasts containing chlorophyll; (2) why a fresh leaf feels rigid and crisp — turgor pressure from the large central vacuole pressing the cell membrane against the rigid cell wall; and (3) why the leaf wilts and dies when left in the sun — the vacuole loses water, turgor is lost, and the cell wall alone cannot maintain the leaf's shape. This explanation directly answers the anchoring phenomenon question and advances the unit's driving question: what is inside the cell, and how does it keep a living thing alive?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Photosynthesis in organisms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens their science notebook and draws a quick sketch (2–3 minutes) of what they predict a sukuma wiki leaf cell looks like under the electron microscope, based on what they already know from Lessons 2 and 3. They label any structures they expect to see and add a written sentence answering: 'How do you think a sukuma wiki plant cell will be the same as — and different from — the human cheek cell we studied in Lesson 3?' Learners then share their prediction with their shoulder partner before the teacher takes cold-calls.	Open by anchoring to the phenomenon: say exactly — 'Remember our Grade 10 learner in Kisumu who saw tiny box-like compartments packed with green granules under the school light microscope, and then saw something far more complex under the electron microscope? Today we are going to look at that electron microscope image together — but first, I want to know what YOU predict is inside that sukuma wiki cell.' Pause. Allow 2 minutes of silent individual sketching. Then say: 'Share your sketch with your shoulder partner. Tell them: one structure you think will be the same as in the cheek cell, and one structure you think will be different or new.' Allow 60 seconds of pair talk. Then cold-call 4 learners (aim for gender balance: 2 female, 2 male) using non-volunteering selection — scan the room and choose learners who have not yet spoken this week. For each response, wait 12 seconds after the question before calling on the next learner. Record 3–4 predicted structures on the board under two columns: 'Same as	Predict–Check anchoring: By making predictions public and writing them on the board before observation, learners activate prior knowledge from Lessons 2 and 3 and create a personal stake in the upcoming evidence. The visible prediction board creates cognitive dissonance when evidence contradicts predictions — a key driver of conceptual change. Recording predictions also gives the teacher instant formative data on residual misconceptions (e.g., learners who predict animal and plant cells are identical, or who do not predict a cell wall).	Teacher scans sketches during pair-share — look for: (1) Whether learners include a nucleus, mitochondria, and cell membrane (carried over from Lesson 3 — expected); (2) Whether any learner predicts a cell wall, chloroplasts, or large vacuole (demonstrates advanced prior knowledge); (3) Whether learners who struggled in Lesson 3 can recall at least 3 animal cell organelles. Flag groups that produce blank or minimal sketches — these groups will need closer monitoring during the Observe phase. Note on a clipboard which learners were cold-called.

		animal cell' and 'New / Different?' Do NOT correct predictions yet — say: 'We are going to keep these predictions visible on the board and check them against our evidence.'		
Observe Phase 📹 **VIDEO:** [Photosynthesis in organisms](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Plant Cell Structures (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Each group of four receives a printed A3 evidence sheet containing two electron micrograph photomicrographs: Image A — a transmission electron micrograph (TEM) of a sukuma wiki (kale) leaf mesophyll cell at low magnification (~5,000×), clearly showing the cell wall, large central vacuole, multiple chloroplasts, mitochondria, nucleus, and cell membrane; Image B — a high-magnification TEM (~50,000×) of a single chloroplast from the same cell, showing the outer double membrane, internal thylakoid membranes arranged in grana, and the stroma. Learners use the Diagram Table worksheet (introduced in Lesson 3) and add a new column: 'Plant Cell (Sukuma Wiki / Onion)'. Working silently for 5 minutes, each learner individually annotates their own copy of the photomicrographs — drawing arrows and labelling every structure they can identify. Groups then compare annotations for 4 minutes and agree on a consensus list. Finally, the class shares findings in a 3-minute whole-class debrief, constructing a shared Venn diagram on the board: 'Animal Cell Only | BOTH | Plant Cell Only'.	Distribute evidence sheets face-down. Say: 'In a moment I am going to ask you to turn over your evidence sheet. You will have 5 minutes of silent individual observation — no talking, no asking me for answers. Your job is to be a scientist: look carefully at every part of both images and label everything you can identify. Use the organelle names from your Lesson 3 notes, and add question marks next to anything you see that you cannot name yet.' Say 'Turn over — begin.' Set a visible 5-minute timer. Circulate — do NOT answer identification questions during individual observation. After 5 minutes say: 'Now compare with your group. You have 4 minutes. Task: agree on your consensus list for the Plant Cell column of your Diagram Table. Pay special attention to anything you see in these plant cell images that was NOT in the human cheek cell images from Lesson 3.' Circulate and listen for group discussion. Prompt struggling groups with: 'Look at the outermost boundary of the cell — what do you notice that is different from the cheek cell?' (WAIT 12 seconds before offering further prompting.) After 4 minutes, bring the class together for the Venn diagram construction. Draw two overlapping circles on the board labelled 'Animal Cell Only' and 'Plant Cell Only' with 'BOTH' in the centre. Cold-call 6 groups in	Evidence-Based Annotation + Collaborative Venn Diagram Construction: The two-stage process (individual silent annotation → group consensus → whole-class Venn diagram) is a structured inquiry routine (NGSS Science and Engineering Practice 4: Analysing and Interpreting Data; SEP 6: Constructing Explanations). Silent individual annotation first ensures every learner engages with the evidence personally before group discussion; the Venn diagram makes structural comparisons visible and spatial, reducing cognitive load when learners hold multiple organelles in mind simultaneously. Returning to the predictions board at the end of the phase creates an explicit predict–check–revise loop, a core metacognitive move.	During individual annotation: teacher scans for learners who leave Image B (chloroplast) entirely unlabelled — this indicates they do not yet connect the green granules from the light microscope to the chloroplast ultrastructure, a key conceptual link to address in the Explain phase. During group work: listen for whether groups spontaneously mention 'turgor', 'photosynthesis', or 'support' in relation to the vacuole, chloroplast, and cell wall — these are marks of conceptual transfer beyond mere naming. During Venn diagram construction: the completed diagram on the board is a class-level formative artefact — photograph it for use in teacher reflection. Key error to watch for: learners placing 'mitochondria' in 'Plant Cell Only' (a common misconception that plants do not respire — address immediately during the Venn construction with: 'Check Image A again — can anyone spot a mitochondrion in the sukuma wiki cell?').

		sequence — each group contributes ONE structure to the diagram. After each contribution, ask the class: 'Do you all agree that this belongs in this section? Hands up if you disagree — tell me why.' WAIT 10 seconds. Probe with: 'Where did we see evidence for this in the photomicrograph — which image, which part of the image?' Ensure the final Venn diagram includes: Animal Only — centrioles, lysosomes (from Lesson 3); Both — nucleus, mitochondria, ribosomes, rough ER, cell membrane, Golgi apparatus; Plant Only — cell wall, chloroplasts, large central vacuole (tonoplast). Say: 'Let us go back to our predictions board. Which of our predictions were correct? Which surprised you?'		
Explain Phase 📹 **VIDEO:** Photosynthesis in organisms Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups receive a second worksheet: 'Structure–Function–Phenomenon Links' — a three-column table with column headers: 'Plant-Specific Structure', 'What it does (Function)', 'How it explains what we SEE in the sukuma wiki leaf'. Groups have 6 minutes to complete the three rows (cell wall, chloroplast, large central vacuole) using their annotated photomicrographs and Diagram Table as evidence. Each group then presents one row to the class in a 2-minute round-robin. The teacher uses their responses to build a class-level explanation connecting each structure to the anchoring phenomenon — the wilting, drying sukuma wiki leaf.	Distribute the Structure–Function–Phenomenon Links worksheet. Explain the task: 'Your job for the next 6 minutes is to explain — not just name. For each of the three plant-only structures, I want you to: describe what it looks like in the electron micrograph, say what job it does inside the cell, and then connect it to something we can actually SEE when we look at a fresh sukuma wiki leaf or a wilted one.' Model one example aloud using a non-plant-specific organelle: 'For example, for mitochondria I would write: looks like a dark oval with folded inner membranes; releases energy from food by respiration; this explains why the leaf stays alive and can grow.' Then say: 'Now your groups do the same for cell wall, chloroplast, and large central vacuole. Use your annotated photomicrographs as evidence.'	Structure–Function–Phenomenon Linking (Causal Explanatory Mapping): This strategy (aligned with NGSS SEP 6: Constructing Explanations and SEP 8: Obtaining, Evaluating, and Communicating Information) requires learners to move through three levels: structural description (what it looks like in the EM), mechanistic function (what it does), and phenomenological connection (what it explains about the visible sukuma wiki leaf). This three-level mapping prevents shallow rote learning of organelle names by anchoring every structure in a causal explanation of the anchoring phenomenon. The teacher's football analogy for the vacuole is a Bridging Analogy — a deliberate pedagogical tool to connect an unfamiliar concept (turgor pressure) to a familiar physical experience.	Collect all completed Structure–Function–Phenomenon worksheets at the end of the phase — these are the primary formative assessment artefact for Lesson 4. Evaluate each group's worksheet against three criteria: (1) Accuracy of function (e.g., chloroplast = photosynthesis, not just 'makes food'); (2) Quality of phenomenon connection (superficial: 'makes the leaf green' vs. deep: 'chlorophyll in the thylakoid membranes absorbs sunlight to drive photosynthesis, which produces the glucose that keeps the leaf alive'); (3) Use of evidence from the photomicrograph (do they cite specific image features?). Groups scoring 0–1 on criterion 3 need targeted feedback before Lesson 5. Listen during group presentations for spontaneous use of the term 'turgor pressure' — if

		Set 6-minute timer. Circulate. Prompt groups that are stuck on 'chloroplast function' with: 'In Lesson 2 we said the green granules absorb sunlight — what process does that power?' WAIT 12 seconds. For groups stuck on the vacuole: 'Think about what happens to a football when you let the air out — now think about the cell.' After 6 minutes, cold-call one group per row (3 different groups). After each group presents, ask the class: 'Does anyone want to add to or challenge what Group ___ said?' WAIT 10 seconds. Then deliver a 90-second teacher synthesis, narrating: 'So let us go back to Kisumu. Our learner sees a fresh sukuma wiki leaf — it is GREEN because every mesophyll cell is packed with chloroplasts containing chlorophyll. It is RIGID and crisp because the large central vacuole is full of water, pressing the cell membrane against the tough cell wall — we call that turgor pressure. But when the leaf sits in the sun for days, what happens? The vacuole loses water. Turgor is lost. The cell wall alone cannot hold the shape. The leaf wilts — and eventually the cells die. Every structure we labelled today explains one part of what our learner in Kisumu saw.'		no group uses it, introduce it explicitly in the teacher synthesis.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Photosynthesis in organisms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner receives two sticky notes. On the first (green), they write one thing they now understand about the driving question ('What makes up an organism — and how do the tiny structures inside a single cell keep a living thing alive?') that they did not understand at the start of the lesson. On the second (orange), they write one new question the	Say: 'Before we update our Driving Question Board, I want you to re-read the driving question at the top of the board.' Read it aloud slowly: 'What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?' Pause 10 seconds. Then say: 'On your green sticky note,	Driving Question Board (DQB) as a Living Knowledge Map: The DQB routine (a core NGSS storyline tool) serves two sensemaking functions simultaneously. The green sticky notes externalise and consolidate new understanding, forcing retrieval practice (a high-efficacy learning strategy). The orange sticky notes make intellectual	Teacher reads all posted sticky notes before the next lesson (takes approximately 5 minutes). Green notes: check for accuracy — any green note with an incorrect claim (e.g., 'I now understand that chloroplasts are found in animal cells too') must be flagged and corrected at the opening of Lesson 5. Orange notes: tally the categories of new questions —

📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	lesson has raised for them — something they observed or discussed that they still cannot fully explain. Learners post both sticky notes on the class Driving Question Board under the column 'Lesson 4: Plant Cell'. The teacher reads 3–4 orange sticky notes aloud and groups similar new questions together on the board, flagging which will be addressed in upcoming lessons.	write one thing you understand NOW that you did not understand at the START of today's lesson — be specific, use organelle names. On your orange sticky note, write one question this lesson has created for you — something you are now curious about or confused by. You have 2 minutes.' After 2 minutes, direct learners to post sticky notes on the DQB. Read 4 orange notes aloud (choose a mix of factual and deeper questions). Say: 'I am going to group these questions. Some of you are asking HOW the chloroplast actually turns sunlight into food — we will address that in our strand on Nutrition. Some of you are asking WHY plant cells need both a cell membrane AND a cell wall — excellent question for our next lesson on cell function. Some of you are asking: if plant cells have mitochondria AND chloroplasts, do they both photosynthesise AND respire? — write that one large on the board — we will come back to it in our Respiration strand.' Point to the DQB and say: 'This board is our growing record of understanding. Every lesson we add to it. Notice how many questions have moved from the orange column to the green column since Lesson 1 — that is your learning, visible.'	uncertainty visible and valued — signalling to learners that scientific curiosity is productive, not a sign of failure. Grouping similar orange questions on the board models the scientific practice of identifying patterns in data (SEP 4) and helps learners see that their individual confusion is often shared, reducing anxiety and building a collaborative inquiry culture.	questions about chloroplast mechanism, questions about cell wall vs. membrane, questions about plant respiration. If more than 40% of orange notes cluster around one topic (e.g., chloroplast function), that topic requires re-teaching or extension in Lesson 5's opening. Orange notes that reveal prior misconceptions (e.g., 'Do plants not need to eat because they make their own food in the mitochondria?') are particularly valuable — note these learners' names for targeted follow-up.
Model Building Phase 📹 **VIDEO:** [Photosynthesis in organisms] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners open their science notebooks to their cumulative cell model — the annotated diagram of the animal cell built in Lesson 3. They now draw a second diagram directly alongside it: a labelled plant cell model based on the electron micrograph evidence gathered today. The plant cell diagram must include at minimum:	Say: 'For the final 5 minutes, you are going to update your personal cumulative cell model. Open your notebook to your Lesson 3 animal cell diagram. Alongside it — leave a 2 cm gap — draw your plant cell model. This is YOUR model, based on YOUR observations today. It should look like what you saw in the electron micrograph,	Cumulative Model Revision (Progressive Modelling): By requiring learners to draw their plant cell model adjacent to their previously constructed animal cell model, this phase enacts the NGSS Science and Engineering Practice 2 (Developing and Using Models) in a cumulative, iterative way. The side-by-side placement	During circulation, teacher uses a 3-point rapid rubric to mentally assess each learner's model: (1) Structural accuracy — are the three plant-specific structures present and correctly labelled? (2) Functional annotation — does each structure have a function note, not just a name? (3) Phenomenon connection — does

📖 **HTML:** [Plant Cell Structures (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	cell wall (with a note: 'rigid, cellulose, provides structural support'), cell membrane (inside the cell wall), large central vacuole (labelled with 'tonoplast' for the surrounding membrane and 'turgor pressure' as the function), at least two chloroplasts (with a sub-label: 'contains chlorophyll — absorbs sunlight for photosynthesis'), nucleus, mitochondria, and ribosomes. Learners annotate each structure with both its function and one connection to the anchoring phenomenon. Finally, learners draw a simple Venn diagram in their notebooks summarising Plant Only / Both / Animal Only, to serve as a personal study reference.	not a textbook cartoon.' Circulate as learners draw. Check that all learners include the three plant-specific structures. Prompt learners who draw chloroplasts as simple circles with: 'Go back to Image B — what do you see INSIDE the chloroplast? Can you add that detail to your model?' (WAIT 12 seconds for learner to look back at the evidence sheet and revise.) Prompt learners who omit mitochondria from their plant cell diagram with: 'Remember our Venn diagram — which structures are in BOTH columns? Does the plant cell have mitochondria?' Close the lesson by saying: 'Your model is now showing two cells side by side — an animal cell and a plant cell. In the next lesson, we are going to push one level deeper: we will look at the FUNCTIONS of these organelles in detail, and we will start to build our explanation of how the whole sukuma wiki leaf — made of millions of these cells — stays alive, grows, and eventually wilts and dies. Take a good look at your model. Every structure on that diagram is doing a job, right now, in every sukuma wiki leaf in every shamba in Kenya.'	physically instantiates the comparison work done during the Observe and Explain phases — making structural differences visible and permanent in the learner's own notebook. The requirement to annotate functions (not just names) and phenomenon connections ensures the model is explanatory, not merely representational. This cumulative model will be revisited and extended in every subsequent lesson, eventually becoming the learner's personal scientific explanation of the entire driving question.	at least one annotation link back to the wilting/green sukuma wiki leaf? Learners whose models score 1 on criteria 2 and 3 (labels only, no functions or connections) receive a written prompt in their notebook margin: 'Add: what does this structure DO, and how does it explain what the leaf looks like?' These learners are prioritised for check-in at the start of Lesson 5. Collect 4–5 notebooks from different groups to review in detail before Lesson 5 — photograph models that exemplify high-quality annotation to share (anonymously) as models at the start of Lesson 5.

D. TEACHER REFLECTION

1. During the Observe phase, did all groups successfully identify the large central vacuole in the sukuma wiki electron micrograph — and did they distinguish it from the cytoplasm? If not, what alternative visual prompt or analogy could help learners 'see' the vacuole's boundary (the tonoplast) more clearly in Lesson 5's opening?

2. When groups presented their Structure–Function–Phenomenon Links, how many connected the chloroplast to photosynthesis using the word 'sunlight' or 'energy' versus simply writing 'makes food'? What does this tell me about learners' readiness for the Nutrition strand?

3. Did any group spontaneously use the term 'turgor pressure' during the Explain phase, or did it emerge only after my football analogy? What does this suggest about the effectiveness of that bridging analogy, and should I use a more locally familiar analogy (e.g., a fully inflated football vs. a deflated one seen at a local pitch in

Kisumu)? 4. Reading the orange sticky notes from the DQB: what is the single most common unresolved question from today's lesson, and how will I address it in the opening 3 minutes of Lesson 5 without derailing the planned lesson structure? 5. When learners drew their cumulative plant cell models, did they place the vacuole correctly (large, central, pushing other organelles to the periphery) or did they draw it as a small circle like a mitochondrion? Incorrectly sized vacuoles suggest the learner has not yet connected vacuole size to turgor pressure — how will I address this in Lesson 5's model review?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Electron micrograph photomicrographs of sukuma wiki leaf mesophyll cells and onion cells at two magnifications, showing: a large central vacuole bounded by the tonoplast; multiple chloroplasts with visible internal thylakoid membranes arranged in grana; a thick cell wall outside the cell membrane; plus shared organelles (nucleus, mitochondria, ribosomes) also seen in animal cells in Lesson 3.
What did I learn?	Plant cells contain three structures absent in animal cells — the cell wall (cellulose, provides rigid structural support), chloroplasts (contain chlorophyll arranged in thylakoid membranes, drive photosynthesis using sunlight), and a large central vacuole (stores water, generates turgor pressure by pressing against the cell wall). All other major organelles (nucleus, mitochondria, ribosomes, cell membrane, rough ER, Golgi apparatus) are shared with animal cells.
How does this explain the phenomenon?	The fresh sukuma wiki leaf is green because mesophyll cells are densely packed with chloroplasts containing chlorophyll. It feels rigid and crisp because the large central vacuole in each cell is full of water, generating turgor pressure against the rigid cell wall — giving the whole leaf mechanical strength. When the leaf is left in the sun without water, the vacuole loses water, turgor pressure drops, and the cell wall alone cannot maintain the leaf's shape — so the leaf wilts, then dries and dies, exactly as the learner in Kisumu observed.

LESSON 5 (40 minutes): Organelle Functions — What Does Each Structure in the Sukuma Wiki Cell Actually Do?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners consolidate their understanding of cell ultrastructure by matching each organelle to its specific function, directly connecting every structure back to the anchoring phenomenon of the wilting and dying sukuma wiki leaf.
Knowledge	– State the specific function of each major organelle in plant and animal cells (nucleus, mitochondria, chloroplasts, ribosomes, cell membrane, cell wall, vacuole, Golgi apparatus, rough endoplasmic reticulum, lysosomes). – Explain how organelle functions are interdependent — for example, how ribosomes, rough ER, and the Golgi apparatus form a protein-secretion pathway. – Distinguish between organelles found only in plant cells (chloroplast, cell wall, large central vacuole) and those found only in animal cells (centrioles, lysosomes) and those shared by both.
Skills	– Analyse function descriptions and match them to correct organelle names using a physical card sort (NGSS SEP 6: Constructing Explanations). – Justify card-sort decisions using evidence from Lessons 3 and 4 observations and communicate reasoning to peers (NGSS SEP 7: Engaging in Argument from Evidence).
Attitudes	– Appreciate that the survival of the sukuma wiki plant — and the health of the Kisumu learner who eats it — depends on a precisely coordinated team of organelles working together inside each cell. – Develop curiosity about how the disruption of even one organelle's function can lead to the wilting and death observed in the anchoring phenomenon.
Key Inquiry Question	What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?
Purpose in Storyline	Lessons 3 and 4 built a detailed structural picture of animal and plant cell organelles through observation of photomicrographs. Lesson 5 is the pivotal Explain Phase: learners now assign a specific job to every structure they identified, completing the link between structure and function. This directly answers why the sukuma wiki leaf dies when cut off from sunlight and water — each organelle stops doing its job — and prepares learners for Lessons 6–8 which explore how organelle functions scale up through tissues, organs, and organ systems.
Safety Notes	No specific hazards. Card-sort materials are paper-based. Ensure learners handle printed cards carefully and return all sets intact for reuse.

B. LESSON OVERVIEW

Lesson 5 is the dedicated Explain Phase of the 8-lesson evidence-gathering sequence on The Cell. After four lessons spent anchoring the phenomenon (a wilting sukuma wiki leaf), comparing microscopes, and building detailed structural knowledge of animal and plant cell ultrastructure, learners are now ready to assign precise biological functions to each organelle they have identified. The lesson centres on a tactile card-sort activity: each pair of learners receives a set of 20 cards — 10 'Function Description' cards written in accessible, Kenya-relevant language (e.g., "This organelle uses sunlight, water from Lake Victoria, and carbon dioxide to make the glucose that sweetens githeri") and 10 'Organelle Name' cards. Pairs sort, debate, and justify their matches before merging into groups of four to resolve disagreements through argument from evidence.

The teacher's role in this lesson shifts from information provider to sensemaking facilitator. After pairs complete their initial sort, the teacher conducts a whole-class reveal using a projected reference diagram, pausing at each organelle to explicitly connect its function back to the anchoring phenomenon. For example, when revealing that chloroplasts carry out photosynthesis, the teacher asks: "If the sukuma wiki leaf is cut and left in the sun without water, the chloroplasts lose the raw material they need — what do you predict happens to the leaf's energy supply, and how does that connect to what we observed in Lesson 1?" These phenomenon connections ensure that function knowledge is never abstract but always grounded in the observable event that opened the unit.

The lesson closes with learners updating the class Driving Question Board (DQB), moving sticky notes from the 'We Wonder' column to the 'We Now Know' column, and adding new questions that have emerged — particularly about what happens when organelles malfunction or when a cell is deprived of water or light. Learners also make the first annotated revision to their cumulative cell model, adding function labels alongside the structural labels they added in Lessons 3 and 4. By the end of 40 minutes, every learner should be able to explain, in their own words, why a sukuma wiki cell that is cut off from sunlight and water will eventually stop functioning and die.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Cell parts and their functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Functions that Describe Situations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Before receiving the card-sort materials, learners open their science notebooks to a T-chart headed 'Organelle' and 'What I think it does.' Working individually for 3 minutes, they fill in as many organelle functions as they can recall from Lessons 3 and 4, using only memory — no notes. They pay particular attention to the three plant-specific organelles (chloroplast, cell wall, large central vacuole) because these are central to the anchoring phenomenon. After individual writing, learners share with a shoulder partner for 1 minute, noting where they agree or disagree.	Start the lesson by projecting the anchoring phenomenon image: the Lesson 1 split-screen showing the wilted sukuma wiki leaf on the left and the electron micrograph with labelled organelles on the right. Say: 'Last lesson we identified all these structures inside the sukuma wiki cell. Today's big question is: what is each one actually doing to keep this plant alive?' Instruct learners: 'On your own, in your notebook, write the name of every organelle you can remember from Lessons 3 and 4, and next to each one write what you think its job is. You have 3 minutes — go.' [Start timer. Circulate silently. Do NOT answer questions about correct answers — redirect with 'Write your best guess for now.'] After 3 minutes, call time and say: 'Share your list with your partner. Where do you disagree, put a question mark — those are the ones we will investigate today.' [WAIT TIME 10 seconds after instruction before expecting sharing to begin.] Cold-call 3 pairs: 'Pair at the front — what was your biggest	Prior Knowledge Activation via T-Chart: By requiring learners to commit their current understanding to writing before any instruction, this strategy surfaces both correct prior knowledge and persistent misconceptions (e.g., confusing the cell wall with the cell membrane, or attributing energy release to the nucleus). The teacher's note of contested organelles on the board transforms individual uncertainty into a class-level inquiry focus, motivating the subsequent card sort.	Circulate and scan T-charts during the 3-minute individual writing phase. Look for: (1) whether learners correctly distinguish chloroplast from mitochondria — a common confusion; (2) whether any learner correctly identifies the ribosome–rough ER–Golgi secretion pathway; (3) whether learners who struggled in Lessons 3–4 can recall at least 5 organelles. Flag learners who have fewer than 4 organelles written down — seat them next to stronger peers during the card sort.

		disagreement?' Note 2–3 contested organelles on the board (likely: Golgi apparatus, ribosomes, rough ER). Say: 'Keep those question marks — the card sort is about to help you resolve them.'		
Observe Phase 📹 **VIDEO:** [Cell parts and their functions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Functions that Describe Situations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each pair receives an envelope containing 20 laminated cards: 10 'Organelle Name' cards (nucleus, mitochondria, chloroplast, ribosome, cell membrane, cell wall, large central vacuole, Golgi apparatus, rough endoplasmic reticulum, lysosome) and 10 'Function Description' cards written in Kenyan context. Examples: 'I use sunlight, carbon dioxide from the Kisumu air, and water to make glucose — the same glucose that ends up in the sukuma wiki you eat for dinner' [chloroplast]; 'I am the gatekeeper — like the entry barrier at a Nairobi supermarket, I decide what enters and leaves the cell' [cell membrane]; 'I am the manager's office — I hold the master blueprint for every protein the cell will ever make' [nucleus]; 'I am the jiko — I break down glucose from ugali and githeri to release ATP energy the cell can use immediately' [mitochondria]; 'I am the postal sorting office at the Kisumu GPO — I package, label, and dispatch proteins to wherever they are needed' [Golgi apparatus]; 'I am the rigid fence around the homestead — I give the sukuma wiki cell its box-like shape and prevent it from bursting when the vacuole fills with water' [cell wall]; 'I am the water tank — when I am full, I press against the fence and keep the sukuma wiki leaf stiff and upright; when I empty, the leaf wilts' [large central	Distribute pre-prepared envelopes. Say: 'Inside your envelope you have 10 organelle name cards and 10 function cards. Your task is to match every function card to the correct organelle name card. You must talk — no silent sorting. Every match needs a reason: say the reason out loud to your partner before you place the card.' [WAIT TIME 12 seconds to allow all pairs to open envelopes and begin reading cards before further instruction.] Set a 7-minute timer. Circulate actively, listening for misconceptions. When you hear a misconception, do NOT correct it immediately. Instead, ask: 'What evidence from the electron micrograph in Lesson 4 supports that match?' or 'Where did you observe that structure — in the plant cell, the animal cell, or both? Does the function card match that?' After 7 minutes, say: 'Now find another pair near you — groups of four. Compare your matches. Where you disagree, use your Lesson 3 and 4 notes to argue for your answer. You have 4 minutes.' Cold-call 2 groups during the group discussion phase: 'Group by the window — which match caused the most disagreement and what evidence settled it?' Record the most contested matches on the board.	Physical Card Sort with Forced Justification: The tactile manipulation of cards externalises reasoning and makes the matching process visible to both learners and teacher. The requirement to verbalise a justification before placing each card activates the NGSS SEP 7 practice (Engaging in Argument from Evidence) at a low-stakes, peer level. The Kenya-contextualised function descriptions (sukuma wiki, Kisumu GPO, jiko, City Council truck) reduce cognitive load by connecting unfamiliar biological vocabulary to familiar local referents, making abstract organelle functions concrete and memorable.	Listen specifically for three high-value misconceptions during circulation: (1) Learners placing 'ribosome' with the function card for Golgi apparatus (confusion about protein synthesis vs. protein packaging); (2) Learners placing 'cell membrane' with the cell wall function card (confusion about rigidity vs. selective permeability); (3) Learners placing 'mitochondria' with the chloroplast function card (confusion between photosynthesis and cellular respiration). Tally how many pairs hold each misconception — use this tally to decide how much time to spend on each organelle during the whole-class reveal in the Explain Phase.

	vacuole]; 'I am the small workshop fundi — I read the blueprint and build every protein in the cell' [ribosome]; 'I am the delivery road from the manager's office to the postal sorting office — proteins travel along my folded membranes' [rough ER]; 'I am the City Council waste truck — I break down old organelles and digest bacteria that sneak into the cell' [lysosome]. Pairs lay all 20 cards on the desk and physically match each function card to an organelle name card, discussing each match aloud. They record their final matches in their notebooks.			
Explain Phase 📹 **VIDEO:** [Cell parts and their functions] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Functions that Describe Situations (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	The teacher projects a colour-coded master diagram of a plant cell (with all 10 organelles labelled by number, not name) alongside a matching table. The teacher reveals the correct match for each organelle one at a time, pausing for learners to check their card-sort result and correct their notebook records. For each organelle, the teacher explicitly connects the function back to the anchoring phenomenon using a 'Function → Phenomenon Link' sentence frame: 'This organelle does [X], which matters for the sukuma wiki leaf because [Y].' Learners record both the function and the phenomenon link in a two-column table in their notebooks. After all 10 matches are revealed, learners answer three consolidation questions in their notebooks: (1) 'Which two organelles must work together so that the sukuma wiki leaf can capture the sun's energy AND release it when needed?' (Ans: chloroplast + mitochondria); (2) 'What happens to the large central vacuole when a sukuma	Project the master diagram. Say: 'We are now going to check your card sorts together. I will reveal one organelle at a time. Before I tell you the answer, I want you to look at your card and hold it up — thumbs up if you are confident, thumbs sideways if you guessed.' [This gives instant whole-class formative data before revealing answers.] Reveal each match in this recommended order (build from familiar to complex): cell membrane → nucleus → ribosome → mitochondria → chloroplast → cell wall → large central vacuole → rough ER → Golgi apparatus → lysosome. For EACH organelle, deliver the phenomenon link. Scripted examples: CHLOROPLAST: 'Chloroplasts carry out photosynthesis — they capture the energy in the sunlight falling on this sukuma wiki leaf in Kisumu and use it to make glucose. When this leaf is cut and left without water, the chloroplasts cannot complete photosynthesis — the glucose supply drops — and the cell begins to starve. That	Phenomenon-Anchored Reveal with Thumbs Formative Check: The sequential reveal with a pre-reveal confidence signal (thumbs up/sideways) turns a traditional teacher-led correction into an active prediction-check cycle, maintaining learner cognitive engagement. By scripting a 'Function → Phenomenon Link' for every organelle, the teacher ensures that structural knowledge never floats free of the anchoring event — every new piece of information is immediately tied to the observable reality of the wilting sukuma wiki leaf. This is consistent with NGSS SEP 6 (Constructing Explanations) because learners are explicitly building a causal chain from organelle function to macroscopic observable outcome.	Collect the thumbs signals before each reveal — count the number of thumbs-sideways responses. If more than one-third of the class is thumbs-sideways for chloroplast, mitochondria, or cell wall, extend the explanation and use an additional analogy. Circulate during the 5-minute consolidation question period and check Question 2 specifically: learners who cannot connect vacuole water loss to turgor pressure and wilting have not yet made the structure-to-phenomenon link. Pull these learners into a brief teacher-led re-explanation before the DQB phase. Target success criterion: at least 80% of learners correctly answer all three consolidation questions.

	wiki leaf is left to wilt in the sun? How does this affect the whole leaf?' (Ans: vacuole loses water → turgor pressure drops → leaf wilts); (3) 'Name ONE organelle that is in the body cells of the learner in Kisumu AND in the sukuma wiki leaf cell — and explain why both need it.' (Many valid answers, e.g., mitochondria — both need ATP energy to carry out life processes.)	is the beginning of the wilting we saw in Lesson 1.' [WAIT TIME 12 seconds. Then cold-call: 'Achieng, what would happen to the chloroplasts if the leaf were also kept in a dark room?'] LARGE CENTRAL VACUOLE: 'This water tank is the reason a fresh sukuma wiki leaf stands upright. When the vacuole is full of water, it pushes outward against the rigid cell wall — we call that turgor pressure. Take away the water — the vacuole deflates — turgor pressure drops — the cell goes limp — the leaf wilts. This is the direct cellular explanation for what we observed on Day 1.' [WAIT TIME 15 seconds. Then cold-call 2 learners: 'Otieno — what is the word for the pressure the vacuole creates? Wanjiku — what happens to that pressure when the plant is not watered?'] After all 10 are revealed, assign the 3 consolidation questions: 'Work individually — 5 minutes — silent.' Circulate and read over shoulders.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Cell parts and their functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Functions that Describe Situations (Flexbook) Source: CK-12 🔍 Search ARES	Learners walk to the class Driving Question Board, which has been divided into three columns: 'We Wonder,' 'We Now Know,' and 'New Questions.' Each learner selects up to two sticky notes from the 'We Wonder' column that they feel Lesson 5 has now answered (e.g., 'What do chloroplasts do?' or 'Why does the leaf go limp when it dries?'). They write a one-sentence answer on the back of the sticky note and move it to 'We Now Know.' They then write at least one new question sparked by today's lesson on a fresh sticky note and add it to 'New Questions' (e.g., 'If mitochondria release energy from glucose AND	Say: 'Before we update the board, look at it for 30 seconds — read the sticky notes in the We Wonder column and decide which ones Lesson 5 has answered.' [WAIT TIME 30 seconds of silent reading.] 'Now move to the board. Take the sticky notes you can answer, write the answer on the back, and move them to We Now Know. Then write a NEW question — something today's lesson made you wonder about — and add it to New Questions. You have 4 minutes.' Circulate and read new questions as they are posted. After 4 minutes, cold-call 3 learners to read their new questions aloud: 'Kamau — read us your new	Driving Question Board Iterative Update: The physical act of moving sticky notes from 'We Wonder' to 'We Now Know' gives learners a tangible, visible record of growing understanding — a metacognitive tool that makes learning progress concrete. The requirement to generate a new question prevents closure and maintains the epistemic momentum of the unit. Connecting the DQB practice to 'how real science works' positions learners as participants in a scientific community rather than passive recipients of facts, which is consistent with NGSS SEP 1 (Asking Questions).	Read the 'We Now Know' sticky notes to verify that learners' written answers are scientifically accurate and use correct organelle terminology. Flag any sticky notes with inaccurate answers and quietly return them to the learner with a prompt: 'Can you revise this using the phenomenon link we discussed today?' Read the 'New Questions' sticky notes to assess the sophistication of learner thinking: surface-level new questions (e.g., 'What colour are mitochondria?') indicate learners are still thinking at the identification level; deeper questions (e.g., 'If the mitochondria stop working in a sukuma wiki cell, would the cell die

for similar readings	chloroplasts make glucose — does the plant ever use its own glucose at night?' or 'What happens to lysosomes when a cell in my body gets infected by bacteria?'). The class DQB is photographed by the class monitor for the ARES platform record.	question.' After each question, ask the class: 'Who has a partial answer to that already, from any of our lessons?' [WAIT TIME 10 seconds each time.] Capture the most intellectually rich new questions by underlining or starring them on the board — these will seed the DQB for Lessons 6–8. Say: 'Notice how our board is shifting — we started this unit with mostly We Wonder notes. Now we are building up the We Now Know column. But each answer is generating new, deeper questions — that is exactly how real science works in a lab in Nairobi or at a research station in Kericho.'		faster than if the chloroplasts stopped?') indicate learners are thinking at the functional-interdependence level, which is the target for SLO (d).
Model Building Phase 📹 **VIDEO:** Cell parts and their functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Functions that Describe Situations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the cumulative plant cell model diagram they started in Lesson 4 (a hand-drawn or printed outline of a plant cell with organelles labelled by name). Using a different coloured pen or pencil (to make additions visible as a new layer), learners add a small function annotation next to each organelle label — a short phrase of no more than 6 words summarising the function (e.g., next to 'chloroplast' they write 'makes glucose using sunlight'; next to 'large central vacuole' they write 'stores water → turgor pressure → leaf upright'). They also add one 'phenomenon arrow': a bold arrow from the large central vacuole annotation to a small sketch of a wilted leaf in the corner of their diagram, labelled 'when vacuole empties → leaf wilts (Lesson 1 phenomenon).' This visual link physically embeds the phenomenon into the model. Learners who finish early add a second phenomenon arrow	Say: 'Take out the plant cell model diagram you built in Lessons 3 and 4. Pick up a different coloured pen — we are adding a new layer today: function labels. For every organelle on your diagram, write a maximum 6-word phrase describing its job. Six words — no more — make it sharp and precise.' [Project a model example on the board showing the format: organelle name + short function phrase, with a phenomenon arrow to a wilted leaf sketch.] 'You also need to draw at least one phenomenon arrow — connecting an organelle function directly to something we observed in Lesson 1: the wilting sukuma wiki leaf.' [WAIT TIME 10 seconds for learners to locate their diagrams and pens before starting.] Set a 5-minute timer. At the 3-minute mark, say: 'You should have at least 5 function labels by now — if you have fewer than 3, focus on the three plant-specific organelles first: chloroplast, cell wall, vacuole	Cumulative Annotated Model Revision: Adding a visually distinct layer (different colour pen) to an existing model makes the progression of understanding tangible — learners can literally see what they knew after Lesson 4 (structure labels) versus what they know after Lesson 5 (function labels). The 6-word constraint forces precision and prevents copying lengthy definitions without understanding. The mandated 'phenomenon arrow' ensures that model-building is not an isolated academic exercise but a tool for explaining the real-world event that opened the unit, consistent with NGSS SEP 2 (Developing and Using Models).	Collect or photograph 6–8 learner model diagrams at the end of the lesson (rotate which learners each lesson to build a full class sample over the unit). Assess against two criteria: (1) Scientific accuracy — are the function phrases correct? Common errors to look for: 'cell wall makes energy' (confusion with mitochondria), 'vacuole does photosynthesis' (confusion with chloroplast); (2) Phenomenon connection — does the learner's phenomenon arrow correctly link an organelle function to an observable outcome from Lesson 1? Learners who draw the arrow but cannot explain the causal chain verbally when asked need additional support before Lesson 6.

	connecting chloroplast to a sketch of a healthy green sukuma wiki leaf labelled 'photosynthesis keeps leaf alive and green.'	— those are the ones most directly connected to our phenomenon.' After 5 minutes, cold-call 2 learners to share their phenomenon arrow annotation: 'Adhiambo — read us your phenomenon arrow label.' After each sharing, ask: 'Does anyone want to add to what Adhiambo said?' [WAIT TIME 10 seconds.] Close by saying: 'Your model now has both structure — from Lessons 3 and 4 — AND function — from today. In Lessons 6, 7, and 8, we will zoom out: how do cells that all do the same function group together, and how do those groups build up to the whole sukuma wiki plant — or the whole human being in Nairobi?'		

D. TEACHER REFLECTION

1. During the card-sort Observe Phase, which organelle function card caused the most disagreement across pairs? Was it the same organelle that appeared most on the board's contested-matches list? What does this persistent confusion tell you about how the function was introduced in Lessons 3 or 4 — and how will you address it in Lesson 6?

2. When you delivered the 'Function → Phenomenon Link' for the large central vacuole and turgor pressure, how many learners gave a thumbs-up versus thumbs-sideways? If more than one-third were sideways, was the Kisumu water-tank analogy sufficient, or do you need a physical demonstration (e.g., a limp vs. turgid celery stalk in water) at the start of Lesson 6 to reinforce this concept?

3. Review the 'New Questions' sticky notes added to the DQB today. Do they reveal that learners are thinking about organelle interdependence (deeper), or are they still asking identification-level questions (surface)? How will you use the deeper questions to frame the opening of Lesson 6?

4. Look at the phenomenon arrows learners drew on their cumulative models. What proportion correctly connected the large central vacuole (or chloroplast) to a macroscopic observable outcome from the Lesson 1 phenomenon? For learners who drew the arrow but linked it to an incorrect outcome, what targeted questioning strategy will you use at the start of Lesson 6 to repair the misconception without singling learners out?

5. The Kenya-contextualised function descriptions on the cards (Kisumu GPO, jiko, City Council waste truck) were designed to reduce cognitive load. Did learners use these analogies spontaneously in their justifications during the card sort, or did they revert to memorised textbook phrases? If the analogies were not adopted, consider co-creating new local analogies with the class at the start of Lesson 6 to increase ownership and retention.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed, through the card-sort activity and whole-class reveal, that each of the 10 major organelles in plant and animal cells has a distinct and specific function — ranging from the chloroplast capturing sunlight to make glucose, to the large central vacuole storing water and generating turgor pressure, to lysosomes digesting cellular waste.

What did I learn?	Learners learned the specific function of each major organelle (SLO d) and that organelle functions are interdependent: for example, chloroplasts produce glucose, mitochondria release energy from that glucose, ribosomes use that energy to build proteins, and the rough ER and Golgi apparatus dispatch those proteins — forming a coordinated cellular production line that keeps the sukuma wiki plant alive.
How does this explain the phenomenon?	The wilting and dying of the sukuma wiki leaf observed in Lesson 1 can now be explained at the organelle level: when the leaf is cut off from water, the large central vacuole empties, turgor pressure falls, and the leaf goes limp; when cut off from sunlight, chloroplasts cannot perform photosynthesis, glucose production stops, mitochondria have no substrate to release energy from, and the cell's entire machinery gradually shuts down — ultimately causing the leaf to dry and die.

LESSON 6 (40 minutes): Levels of Cell Organisation — From Sukuma Wiki Leaf Cell to Living Organism	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners construct and present concept maps that link organelle → cell → tissue → organ → organ system → organism using both a sukuma wiki plant and a human body as dual Kenyan examples, thereby explaining how the tiny structures they have studied across Lessons 1–5 ultimately sustain a whole living organism.
Knowledge	– Define and distinguish the six levels of cell organisation: organelle, cell, tissue, organ, organ system, and organism. – Trace the organisational hierarchy in a plant using the example: chloroplast (organelle) → leaf cell → leaf mesophyll tissue → leaf → shoot system → sukuma wiki plant. – Trace the organisational hierarchy in an animal using the example: mitochondrion/myofibril (organelle) → muscle cell → muscle tissue → stomach → digestive system → human being in Nairobi. – Explain that each level of organisation performs functions that the level below cannot perform alone.
Skills	– Construct a labelled concept map showing bidirectional links between all six levels of organisation for two Kenyan organisms. – Use evidence from Lessons 3–5 (organelle structures and functions) to justify the arrows and linking phrases on the concept map. – Deliver and respond to peer feedback on concept maps using scientific vocabulary.
Attitudes	– Appreciate that life depends on coordinated organisation at every scale — from a chloroplast granule inside a sukuma wiki cell to the entire plant a Kisumu family eats for dinner. – Respect peers' contributions during group work and gallery-walk feedback, recognising that collaborative model-building produces richer scientific understanding.
Key Inquiry Question	What makes up an organism — and how do the structures inside a single cell scale up to keep a whole living thing alive?
Purpose in Storyline	Lessons 1–5 built detailed knowledge of individual organelles and their functions in sukuma wiki and animal cells. Lesson 6 is the pivotal Explain Phase lesson: learners now zoom out and assemble all that organelle-level knowledge into the full hierarchy of life, directly answering the driving question by showing that the 'tiny box-like compartments packed with green granules' the Kisumu learner saw under the light microscope are not isolated units — they are one level in an unbroken chain of organisation that sustains every living thing.
Safety Notes	No chemical or biological hazards. Learners use scissors and markers for concept-map construction — remind them to handle scissors safely and to cap markers when not in use to avoid inhalation of fumes in an enclosed classroom.

B. LESSON OVERVIEW

Lesson 6 is the Explain Phase centrepiece of the eight-lesson evidence-gathering sequence. Having spent Lessons 2–5 zooming progressively deeper into the sukuma wiki leaf cell — comparing light and electron microscopes, cataloguing animal and plant cell organelles, and mapping the functional interdependencies of those organelles — learners are now ready to zoom back out and answer the driving question at the organisational level. The lesson opens with a provocative prediction task: learners are asked to arrange six scrambled labels (organelle, cell, tissue, organ, organ system, organism) in the order they think represents increasing complexity, and to predict what connects each level to the next. This activates prior knowledge and surfaces any lingering misconceptions about where organelles sit relative to tissues and organs.

The Observe Phase uses a dual-image anchor: a photomicrograph of sukuma wiki leaf mesophyll tissue (showing multiple cells packed together, each with visible

chloroplasts) alongside a labelled diagram of the human stomach wall showing distinct tissue layers. Learners examine these images in pairs, identifying what they can see at each level and recording observations in their science journals. This provides the concrete visual evidence base from which the concept map will be constructed. The Explain Phase is the heart of the lesson. In groups of four, learners receive a concept-map starter template with the six level labels already printed and space for linking phrases and Kenyan examples. Each group constructs two parallel hierarchies — one for the sukuma wiki plant and one for the human body — using evidence from their Lessons 3–5 notes to justify every link. Groups then participate in a gallery walk, placing sticky-note feedback on two other groups' maps before returning to revise their own. The teacher facilitates whole-class sensemaking by cold-calling groups to defend specific linking phrases, driving home the insight that each level of organisation performs emergent functions impossible at the level below. The lesson closes with a DQB update and a cumulative model revision connecting today's zoom-out to the organelle detail of earlier lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Thin-layer chromatography (TLC) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Levels of Measurement (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a set of six label cards (organelle, cell, tissue, organ, organ system, organism) printed on cut paper strips. Working individually for 3 minutes, learners arrange the strips in order of increasing complexity on their desk and write one sentence in their science journal predicting what 'connects' each level to the next — for example, what turns many cells into a tissue. Learners then share their arrangement with a shoulder partner and note any differences before the teacher calls the class together.	Distribute label-card sets as learners settle. Say: 'Before we build anything today, I want to know what YOU already think. Arrange these six labels from smallest to most complex — and for each arrow between levels, write ONE sentence: what connects them?' Allow 3 minutes of individual silent work — do NOT circulate or hint during this time. After 3 minutes say: 'Turn to your shoulder partner. Do your arrangements match? Where do you disagree?' Allow 90 seconds of pair talk. Then cold-call 3 pairs (choose one high-confidence, one uncertain, one who disagreed with their partner): 'Achieng' and Otieno — where did you place organelle relative to cell, and why?' WAIT 12 seconds after each question before accepting responses. Record the two or three most common arrangements on the board without evaluating them. Say: 'We are going to test these predictions today — keep your arrangement on your desk so you can revise it.'	Predict-then-Revise (Elicitation of Prior Knowledge): By committing to an arrangement before instruction, learners activate schema from Lessons 3–5 and create a cognitive hook for the new organisational framework. Recording predictions on the desk makes revision visible and metacognitive — learners can literally see what they changed and why.	Teacher scans arrangements during cold-call: look for whether learners correctly place organelle below cell (a common error is placing 'cell' at the bottom or conflating 'organ' with 'organelle'). Note any learner who cannot place all six — this learner will need targeted support during group work. Record names of learners with incorrect arrangements on a clipboard for follow-up.
Observe	The teacher projects two side-by-	Project or distribute images. Say:	Observation-before-Explanation	Listen for whether learners

Phase 📹 **VIDEO:** Thin-layer chromatography (TLC) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Levels of Measurement (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	side images on the board or distributes printed A4 sheets: (Left) a photomicrograph of sukuma wiki (kale) leaf mesophyll tissue at 400× magnification, showing multiple cells with visible chloroplasts and cell walls packed together; (Right) a labelled cross-section diagram of the human stomach wall, showing mucosa, submucosa, muscle tissue, and serosa layers. In pairs, learners spend 4 minutes examining both images, recording in their journals: (a) what they can identify at the cell level, (b) what they see at the tissue level, and (c) what organ each image belongs to. Pairs share one observation each in a brief whole-class round.	'These two images are your evidence for today's concept map. In your journal, write three rows: Cell level — what do I see? Tissue level — what do I see? Organ — which organ is this part of?' Allow 4 minutes of paired observation. Circulate and listen — do NOT answer questions about organisation yet; redirect with: 'What do you SEE — describe it first, explain later.' After 4 minutes, rapid round: cold-call 4 pairs (2 per image) — 'Kamau and Wanjiru — what did you notice at the TISSUE level in the sukuma wiki image?' WAIT 10 seconds. For the stomach image ask: 'Omondi and Nafula — can you see more than one type of tissue in this cross-section? How do you know they are different tissues and not just different cells?' WAIT 12 seconds. Annotate the projected images with learner responses using a marker or digital annotation tool. Conclude: 'You have just described two levels of organisation from real images. Now your job is to build the full chain — all six levels — for BOTH organisms.'	(Evidence-First Anchoring): Presenting photographic evidence before the concept-map task ensures learners ground their hierarchical model in real observable data rather than rote memorisation. The dual-organism format (sukuma wiki + human stomach) mirrors the two examples they will use in the Explain Phase, creating a seamless scaffold.	correctly identify chloroplasts as organelles within cells within tissue. A key misconception to catch: learners describing the entire leaf as a 'cell' rather than as an organ made of tissues. If this appears, note it for targeted questioning during the Explain Phase. Also check that learners can name the stomach as an organ — not just 'a body part.'
Explain Phase 📹 **VIDEO:** Thin-layer chromatography (TLC) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Levels of Measurement (Flexbook) Source: CK-12	Groups of four receive a concept-map starter template (A3 paper) with six labelled boxes arranged vertically: Organelle → Cell → Tissue → Organ → Organ System → Organism. The template has two columns — one headed 'Sukuma Wiki Plant (Kisumu garden)' and one headed 'Human Body (Nairobi)'. Groups spend 12 minutes filling in: (a) a specific Kenyan example at each level for both organisms (e.g., chloroplast / mitochondrion → leaf cell / muscle cell → mesophyll tissue / muscle	Distribute A3 templates, coloured markers, and sticky notes. Say: 'You have 12 minutes. Every box must have a Kenyan example — not just a generic label. Every arrow must have a sentence that says HOW one level builds the next. And for each step up, name ONE thing the higher level can do that a single cell below it cannot.' Set a visible 12-minute timer. Circulate continuously — prioritise groups that struggled in the Predict Phase. When you see a group writing 'organ system = circulatory	Collaborative Concept Mapping with Gallery Walk Peer Feedback (Developing and Using Models — NGSS SEP 2): Concept mapping is a proven sensemaking strategy that externalises relational thinking, forcing learners to articulate not just WHAT the levels are but HOW they connect. The gallery walk adds a peer-review layer that mirrors scientific practice: scientists present models, receive critique, and revise. The dual-organism format prevents rote copying and	Collect all A3 concept maps at the end of the lesson. Assess against three criteria: (1) All six levels correctly ordered with specific Kenyan examples in both columns — target: 100% of groups. (2) Linking phrases use causal language ('form', 'build', 'work together to') rather than just arrows — target: at least 4 of 6 groups. (3) Emergent function identified at each level transition (the function the higher level performs that the lower cannot) — target: at least 3 of 6 transitions correctly identified

🔍 Search ARES for similar readings	tissue → leaf / stomach → shoot system / digestive system → sukuma wiki plant / human being), (b) a linking phrase on each arrow explaining HOW the lower level builds the higher (e.g., 'Many leaf cells with aligned chloroplasts form mesophyll tissue, which maximises light capture for photosynthesis'), and (c) one function that ONLY the higher level can perform that the lower level cannot. After group construction, the class does a 5-minute gallery walk: each group posts their map on the wall, and every learner places one green sticky-note (something scientifically correct and specific) and one yellow sticky-note (a question or suggested improvement) on two other groups' maps. Groups then return to their own map for 2 minutes to revise based on feedback. The teacher facilitates a 5-minute whole-class debrief, cold-calling groups to defend specific linking phrases.	system' without a linking phrase, prompt: 'Great example — but what does the digestive SYSTEM do that the stomach organ alone cannot do? Think about what happens to food between the mouth and the small intestine.' During gallery walk, say: 'Green note = one thing they got specifically RIGHT. Yellow note = one genuine scientific question you have about their map — not just spelling.' After gallery walk and revision, cold-call three groups for whole-class debrief: (1) 'Akinyi's group — read us your linking phrase between tissue and organ for the sukuma wiki plant.' WAIT 10 seconds. (2) 'Mwangi's group — what function did you say the digestive SYSTEM performs that the stomach organ alone cannot?' WAIT 12 seconds. (3) 'Anyango's group — where do the organelles from Lesson 5 appear on your map, and which organelle is most important at the cell level for keeping the sukuma wiki leaf alive? Defend your answer.' WAIT 15 seconds. Close with: 'Every level on that map is doing something the level below cannot do alone. That is what organisation means in biology — and that is how a chloroplast inside a tiny cell in a sukuma wiki leaf in Kisumu becomes part of a living plant that feeds a family in Nairobi.'	demands genuine transfer of the organisational principle to two different contexts.	per group. During gallery walk, listen for sticky-note quality: yellow notes that ask genuine biological questions (not cosmetic comments) indicate deeper engagement.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Thin-layer chromatography (TLC) Source: Khan	The teacher directs attention to the class Driving Question Board displayed at the front. Learners individually write on a sticky note one sentence completing the prompt: 'I can now explain part of the driving question because I know that…' and post it in the 'Learned Today' column of the	Direct attention to the DQB: 'Look at the Still Wondering column. Read the questions we have posted since Lesson 1.' Give 60 seconds of silent reading. Then say: 'Which of those questions did TODAY'S lesson help answer? Put your hand up if you think we can move a question to the Answered	Driving Question Board — Metacognitive Progress Tracking: The DQB makes the cumulative knowledge-building of the eight-lesson sequence visible to learners. By physically moving questions from 'Still Wondering' to 'Answered', learners experience their own intellectual progress as a	Read all 'Learned Today' sticky notes before the next lesson. Look for notes that correctly name at least two levels and use causal language. Flag any notes that are vague ('I learned about cells') or incorrect ('tissues are inside organelles') — these learners need targeted re-teaching at the start of

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Levels of Measurement (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	DQB. The class then reviews the existing 'Still Wondering' column from Lessons 1–5 and, as a whole group, agrees on which previously posted questions have now been answered by today's concept-map work. Learners may also add new 'Still Wondering' questions prompted by the gallery walk — for example, 'How do the cells in different tissues stay coordinated?' or 'What tells a cell whether to become a leaf cell or a root cell?'	column.' Cold-call 2 learners to nominate questions and justify: 'Chebet — which question would you move, and what did we learn today that answers it?' WAIT 10 seconds. Physically move the sticky note. Then say: 'Now write your OWN sticky note starting with the words: I can now explain… because I know that… Post it in the Learned Today column.' Allow 90 seconds of individual writing. Finally: 'If today's gallery walk raised a NEW question in your mind — something you are still wondering — post it in the Still Wondering column.' Allow 60 seconds. Point to the DQB and say: 'Look at how much of the driving question we can now answer. We started with a Grade 10 learner in Kisumu wondering what those tiny box-like compartments are. We now know what they are, what is inside them, AND how they fit into the chain that keeps the whole plant — and the learner who eats it — alive. Two lessons remain. What is still missing from our explanation?'	concrete, motivating artefact. The closing prompt ('What is still missing?') sustains epistemic curiosity and previews Lessons 7–8.	Lesson 7. Count the number of new 'Still Wondering' questions: if fewer than 3 are posted, the class may not have engaged deeply enough with the gallery walk, signalling a need to revisit peer-feedback quality in Lesson 7.
Model Building Phase 📹 **VIDEO:** Thin-layer chromatography (TLC) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Levels of Measurement (Flexbook) Source: CK-12	In the final 4 minutes, learners open their cumulative science journal model — the annotated cell diagram they have been building and revising since Lesson 2. They add a new outer frame around their existing cell drawing: a bracket labelled 'TISSUE (many cells like this)', a second larger bracket labelled 'ORGAN (e.g., leaf / stomach)', a third labelled 'ORGAN SYSTEM (e.g., shoot system / digestive system)', and a final label 'ORGANISM (sukuma wiki plant / human being in Nairobi)'. Inside the cell, learners draw a small star next to the	Say: 'Open your cumulative journal model from Lesson 2. You have 4 minutes to add the zoom-out layer we built today.' Project a sample completed model on the board showing the nested bracket structure. Say: 'Add the four outer brackets. Label each one with the level AND a Kenyan example from YOUR concept map — not from the board.' Circulate rapidly, checking that learners are labelling with Kenyan examples (sukuma wiki / human being) and not generic terms. At the 3-minute mark say: 'Draw a star on the organelle you argued was most	Cumulative Model Revision — Zoom-Out Extension (NGSS SEP 2: Developing and Using Models): Each lesson in this sequence asks learners to revise the same core model rather than start a new one. This cumulative revision strategy mirrors how scientists iteratively refine models as new evidence emerges. Today's zoom-out extension makes the model structurally complete: learners can now trace a continuous visual path from a single organelle to an entire organism within one annotated diagram.	During the 4-minute model revision, circulate and look for: (1) All four outer bracket levels present and labelled with Kenyan examples — not just copied from the board. (2) The star is placed on an organelle (not a cellular structure like the cell wall) and the justification sentence uses functional language ('because it produces the glucose that powers all cellular work'). (3) The model remains connected to the phenomenon — the sukuma wiki leaf cell should still be identifiable as the core of the diagram. Any learner who cannot add the outer

🔍 Search ARES for similar readings	organelle they argued was 'most important at the cell level' during the Explain Phase debrief, and write a one-sentence justification next to it. This revision zooms the model from organelle-level detail (Lessons 3–5) to organism-level context (Lesson 6), making the full hierarchy visible in a single annotated diagram.	important for keeping the cell alive — and write ONE sentence next to it defending your choice.' At 4 minutes, close the model work: 'Your model now shows the full picture — from the chloroplast granule that Akinyi in Kisumu saw under the microscope, all the way out to the living plant she takes home for dinner. In Lessons 7 and 8, we will look at what happens when parts of this system break down — and what that tells us about how life really works.' Collect journals if time allows for a spot-check, or ask learners to leave journals open on desks for a rapid walk-by scan.		brackets without help is a priority for a 2-minute one-to-one check at the start of Lesson 7.

D. TEACHER REFLECTION
1. During the gallery walk, did learners write genuinely scientific yellow sticky-note questions (e.g., 'How does the organ system coordinate between organs?') or surface-level comments (e.g., 'Your handwriting is messy')? What does this tell me about the depth of their engagement with the concept maps, and how will I improve the gallery-walk protocol in Lesson 7?\n2. Which of the six levels of organisation caused the most confusion — and was it consistent across groups? Common trouble spots are: (a) conflating 'organelle' with 'organ', (b) not being able to name a specific tissue type, or (c) failing to distinguish between a shoot system and a whole plant organism. How will I address this in the Lesson 7 opener?\n3. Were the Kenyan examples (sukuma wiki plant and human digestive system) generative — did they help learners construct the hierarchy — or did they create confusion (e.g., learners unsure which organ system the leaf belongs to)? Would a third Kenyan example (e.g., a tilapia from Lake Victoria) have strengthened or diluted the lesson?\n4. When I cold-called groups to defend their linking phrases, did learners use causal language ('many cells working together form a tissue that can…') or did they revert to listing ('tissue is made of cells')? What targeted questioning strategy will I use in Lesson 7 to push learners from listing to causal explanation?\n5. Looking at the 'Learned Today' sticky notes on the DQB, can I see a clear progression from Lesson 1's wonder ('what are those box-like compartments?') to Lesson 6's understanding ('those compartments are cells, which are one level in a six-level hierarchy that sustains the whole plant')? If not, what aspect of the storyline thread needs to be made more explicit in the remaining two lessons?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a photomicrograph of sukuma wiki leaf mesophyll tissue showing multiple cells with chloroplasts packed together, and a cross-section diagram of the human stomach wall showing distinct tissue layers — providing visual evidence that cells are not isolated units but are organised into tissues and organs.
What did I learn?	Learners constructed concept maps showing the six levels of cell organisation (organelle → cell → tissue → organ → organ system → organism), using two Kenyan examples: the sukuma wiki plant (chloroplast → leaf cell → mesophyll tissue → leaf → shoot system → plant) and the human body (mitochondrion → muscle cell → muscle tissue → stomach → digestive system → human being). They learned that each level performs emergent functions that the level below cannot perform alone.
How does this explain the	The tiny box-like compartments packed with green granules that the Kisumu learner saw under the light microscope are leaf cells

phenomenon?	— one level in an unbroken organisational chain that scales from chloroplast organelles all the way up to the living sukuma wiki plant. This organisation is what keeps the plant — and the human being who eats it — alive: no single organelle or cell could sustain life alone; it is the coordinated hierarchy of tissues, organs, and organ systems that makes a whole organism possible.

LESSON 7 (80 minutes (2 × 40-minute lessons)): 3D Cell Model Project — Building and Justifying a Model of a Plant or Animal Cell

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners consolidate their understanding of cell organelles and their functions by constructing a 3D physical model of either a plant or animal cell using locally available materials, labelling each organelle, and justifying every modelling choice — thereby demonstrating SLO (f): appreciate the function of the cell as a basic unit of life, and SLO (b/c/d): describe and compare plant and animal cell structures and explain the functions of their components.
Knowledge	Learners know the names, locations, and functions of the major organelles of plant cells (cell wall, cell membrane, nucleus, chloroplasts, mitochondria, vacuole, endoplasmic reticulum, ribosomes, Golgi apparatus) and animal cells (cell membrane, nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes, centrioles) as revealed under the electron microscope. They also know the levels of cell organisation: organelles → cells → tissues → organs → organ systems → organism.
Skills	Learners are able to: (1) select appropriate locally available materials to represent each organelle based on its physical characteristics and function; (2) construct and assemble a 3D cell model; (3) label organelles and write their functions on tags; (4) articulate and defend the reasoning behind each material choice using scientific evidence gathered in Lessons 1–6; (5) revise the class Driving Question Board (DQB) by identifying which questions have now been answered.
Attitudes	Learners appreciate the cell as the fundamental unit of life — recognising that the organelles working together inside a single sukuma wiki cell in Kisumu are the same machinery that keeps any living organism, including themselves, alive. They value creativity, precision, and collaborative respect when making and evaluating models.
Key Inquiry Question	What makes up an organism?
Purpose in Storyline	In Lessons 1–6, learners moved from the puzzling observation of the wilting sukuma wiki leaf through microscopy comparisons, organelle identification, and function explanation. Lesson 7 is the synthesis lesson: learners now translate all accumulated evidence into a tangible 3D model. The model externalises their mental model, makes their understanding visible and testable, and directly answers the anchoring phenomenon — every box-like compartment, every green granule, every folded membrane they observed has a name, a material representation, and a function tag. The DQB revisit signals to learners that scientific understanding is cumulative and that building a model is itself a form of scientific argument.
Safety Notes	Learners should handle scissors and any sharp tools carefully; teacher must remind learners to manage waste (polythene off-cuts, clay scraps) responsibly and dispose of it in designated bins to avoid environmental pollution (Environmental Conservation PCI). No food items used in modelling (e.g., maize grains, bean seeds) should be consumed during the lesson. Wash hands after handling clay. Social Justice value: teacher ensures equitable distribution of materials across groups regardless of gender.

B. LESSON OVERVIEW

Lesson 7 is the Model Building Phase of the 8-lesson evidence-gathering sequence for Sub-Strand 1.2: The Cell. Learners work individually or in pairs to construct a 3D physical model of either a plant cell or an animal cell using locally available materials: bottle tops (Golgi vesicles / lysosomes), polythene bags (cell membrane / vacuole membrane), maize grains (ribosomes), bean seeds (mitochondria or nucleus), clay (cytoplasm matrix / cell wall), string (endoplasmic reticulum networks), and cardboard (cell wall frame for plant cell). Each organelle is tagged with its name and function written by the learner. The teacher circulates continuously, asking each learner to justify one modelling choice with scientific reasoning drawn from prior lessons. In the final 15 minutes, learners return to the class Driving Question Board (DQB) to mark questions that their model now answers, and to identify any questions still outstanding. The lesson foregrounds NGSS SEP: Developing and Using Models, and the KICD core competency of Creativity and Imagination, while reinforcing the SLO (f) appreciation of the cell as the basic unit of life connecting a sukuma wiki leaf in Kisumu to the human body of the learner who eats it.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Cells and their Organelles - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a 'Material–Organelle Match' planning sheet listing 10 locally available materials (bottle tops, polythene bag, maize grains, bean seeds, red clay, white clay, string, cardboard, cling film, a small balloon). Working individually for 8 minutes, learners fill in the planning sheet by predicting which material they will use for each organelle and writing one sentence justifying each choice based on the organelle's structure or function. Example prompt on sheet: 'I will use ___ for the chloroplast because ___.' Learners who choose to model an animal cell annotate which organelles they are omitting and why. This activates prior knowledge from Lessons 1–6 and requires learners to connect structure to function before they build.	Distribute planning sheets and materials inventory list. Say: 'Before you touch any materials, I want you to think like a scientist — or like engineer Wangari Maathai planning before she planted. Fill in your planning sheet. You have 8 minutes.' [Start timer visibly.] Circulate silently for the first 3 minutes — do NOT answer questions, let learners wrestle. After 3 minutes, prompt struggling learners with: 'Look back at the electron microscope photomicrograph from Lesson 3 — what shape does that organelle have? Now look at your materials.' After 8 minutes, cold-call 3 learners (ensure gender balance): 'Amina, which material did you choose for the mitochondrion and why?' [WAIT 12 seconds after question before accepting answer.] 'Omondi, do you agree with Amina's choice? Why or why not?' [WAIT 10 seconds.] 'Chebet, you chose a plant cell — what material represents your vacuole and what is your reason?' [WAIT 12 seconds.] Record 3–4 key predictions on the board under the heading: 'OUR PREDICTIONS — We will see if the model supports these choices.'	Anticipation Planning (a structured prediction tool): by committing material choices to paper before building, learners must retrieve and apply their organelle knowledge rather than build randomly. The planning sheet creates a testable commitment — learners can later compare their finished model to their plan, identifying where their understanding was accurate or incomplete. This mirrors how scientists develop hypotheses before an experiment.	Teacher scans planning sheets during circulation. Observable indicator of understanding: learner's justification sentence explicitly names a structural or functional property (e.g., 'I use a polythene bag for the vacuole because it is flexible, holds water, and is semi-transparent, like the tonoplast membrane'). Red flag: learner writes only 'it looks like it' with no functional reasoning — teacher prompts: 'What does that organelle DO? Find the function in your notes from Lesson 4 and use that to justify your choice.'
Observe Phase 📹 **VIDEO:**	Before building begins, learners spend 8 minutes in a focused 'Evidence Review' using the ARES	Direct learners: 'Open the ARES lesson — go to the Observe tab and find the electron microscope	Evidence-to-Model Bridging: learners explicitly return to photographic evidence (the	Teacher observes whether learners' material selections are consistent with their planning

Cells and their Organelles - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	platform: they re-examine the annotated electron microscope photomicrograph of the sukuma wiki leaf cell (plant cell reference) and the animal cell photomicrograph (used in Lesson 3) displayed on their devices. Alongside this, a physical 'Materials Table' at the front of the classroom displays all available construction materials labelled with their names. Learners walk to the Materials Table, select their materials according to their planning sheet, and return to their workspace. As they collect materials, they note on their planning sheet any changes they want to make based on the photomicrograph review — using a different colour pen to show revised thinking.	images from Lesson 3. You have 5 minutes to look carefully at the images one more time before you build. Use this as your blueprint — just like a fundi (artisan) in Nairobi uses a blueprint before constructing a building.' [Set 5-minute timer.] After 5 minutes, say: 'Now, one learner at a time from each row, collect your materials from the Materials Table. Take only what is on your plan.' While learners collect materials, teacher stands at the Materials Table and asks 2 targeted questions to learners as they collect: (1) 'You are picking up string — which organelle is that for, and how does string capture the structure of the endoplasmic reticulum?' [WAIT 10 seconds.] (2) 'You chose bean seeds for the nucleus — what feature of a bean seed reminds you of the nucleus?' [WAIT 12 seconds.] Record any notable revised choices on the board under: 'REVISED PREDICTIONS — New evidence changed our thinking.'	electron microscope images that anchored the phenomenon) immediately before building. This prevents models from becoming purely creative art projects — the photomicrograph acts as a scientific constraint. The 'revised thinking' annotation in a different pen colour makes the process of updating a model based on evidence visible and valued, mirroring how scientists revise models when new data emerges.	sheet justifications and whether revisions are scientifically motivated. Observable indicator: learner picks up a material, pauses, looks at the photomicrograph, and voluntarily swaps it for a different material with a verbal or written reason. Red flag: learner selects materials arbitrarily without reference to the photomicrograph — teacher intervenes: 'Show me on the screen which organelle you are trying to represent. Now describe its shape. Which material on the table best matches that shape AND that function?'
Explain Phase 📹 **VIDEO:** Cells and their Organelles - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners construct their 3D cell model over 30 minutes. They assemble components, attach labels (organelle name + function written on small cardboard tags), and position each organelle in a spatially appropriate location relative to others (e.g., ribosomes attached to the endoplasmic reticulum string; chloroplasts — represented by green-painted maize grains — distributed in the cytoplasm clay; nucleus — bean seed — near the centre). As they build, each learner writes a 'Function Tag' for every organelle: a two-sentence tag stating (1) the organelle's name and (2) what it	Signal the start of building: 'Scientists, you have your blueprint, you have your materials, you have your evidence. You have 30 minutes to construct your model. As you build, write your function tags — every organelle must have one before I visit you.' Circulate continuously. Visit every learner at least once during the 30 minutes. At each visit, point to ONE organelle and ask: 'Justify this choice to me — why did you use THIS material for THAT organelle?' [WAIT 12 seconds after question — do not fill the silence.] Accept and probe responses: 'You said the polythene	Model Construction with Justified Representation (NGSS SEP: Developing and Using Models): the act of physically placing an organelle in a 3D space forces learners to think about spatial relationships between organelles (e.g., ribosomes on rough ER; chloroplasts only in plant cells near the cell membrane for light access). The Gallery Rotate introduces Peer Critique — a strategy where learners evaluate each other's explanatory claims using dot stickers, promoting scientific argumentation without verbal confrontation. Function Tags require learners to translate	Teacher uses a 3-column circulation checklist (Learner name | Organelle justification quality: Structural only / Structural + Functional / Structural + Functional + Phenomenon-linked | Follow-up needed Y/N). Observable indicator of full understanding: learner's justification for a material choice references (a) the physical appearance of the organelle, (b) its function, AND (c) a connection to the phenomenon (e.g., 'I used green-painted maize grains for the chloroplasts because they are small, numerous, and green — just like the green granules we saw in the light microscope view of the

	does in the context of keeping the sukuma wiki plant alive or keeping the learner's own body alive. At the 20-minute mark, learners pause and do a 'Gallery Rotate': they stand and walk to the model on their left, read the function tags, and place a green dot sticker on any tag they agree is accurate and a yellow dot sticker on any tag where they are unsure. Learners return to their own model, read the stickers, and have 5 minutes to revise any tags before the teacher justification conversation.	bag is like the cell membrane because it is flexible — can you add one more reason connected to what the cell membrane DOES?' After Gallery Rotate at the 20-minute mark, say: 'You have 5 minutes — look at the yellow dot stickers on your model. If a peer was unsure, you have two choices: defend your tag using evidence from your notes, or revise it. Either is good science.' During the final 5 minutes of building, cold-call 2 learners to share one function tag aloud to the class: 'Otieno, read your mitochondrion tag and tell us how that connects to the sukuma wiki leaf dying in the sun.' [WAIT 10 seconds.] 'Fatuma, read your cell wall tag — how does your cardboard choice capture what the cell wall actually does?' [WAIT 12 seconds.]	structural knowledge into functional, phenomenon-connected language, deepening sensemaking.	sukuma wiki leaf in Lesson 1, and they capture sunlight to make food that keeps the plant alive'). Red flag: function tags say only 'controls the cell' for the nucleus without explaining how — teacher asks: 'What specifically does the nucleus control, and what would happen to the sukuma wiki cell if the nucleus was removed?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Cells and their Organelles - Example 1] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	With models complete, learners gather as a class in front of the DQB (the shared board of questions accumulated since Lesson 1). Each learner receives two sticky note colours: BLUE ('My model answers this question') and ORANGE ('This question is still open'). Learners spend 6 minutes silently reading every question on the DQB and placing their coloured sticky notes. Then, in a 5-minute whole-class discussion, the teacher facilitates agreement on which questions are now answered and which remain, moving answered questions to a 'EXPLAINED' column and retaining unanswered ones in the 'STILL WONDERING' column. Learners update their personal science notebooks: they write 2–3 sentences explaining how their model answers the original	Direct learners: 'Bring your model and come to the DQB. You have 6 minutes — read every question, and use your blue and orange sticky notes. Think: does your MODEL — the one you just built — provide evidence that answers this question?' [Set 6-minute timer. Remain silent during this period — let learners read and place stickers independently.] After 6 minutes, facilitate discussion by pointing to questions with the most blue stickers: 'Most of us think our model answers this question. Jabali, how does your model answer: What are the green granules in the sukuma wiki cell?' [WAIT 12 seconds.] 'Akinyi, do you agree? Does your animal cell model have the same answer, or is it different?' [WAIT 10 seconds.] Move answered questions to the EXPLAINED column. Point to	Driving Question Board Curation (a metacognitive tracking strategy): the DQB makes the progression of understanding visible over time — learners literally see questions migrating from WONDERING to EXPLAINED. The blue/orange sticky note protocol ensures every learner independently evaluates the DQB before group discussion, preventing dominant voices from steering the class. The notebook writing task requires learners to construct a personal, phenomenon-linked narrative — the highest level of sensemaking in this framework.	Teacher collects notebook entries at the end of the lesson. Observable indicator: entry explicitly names at least two organelles, states their functions, and connects the organelle's role to what happens when the sukuma wiki wilts (e.g., 'If the mitochondria stop working, the cell cannot release energy from glucose, so the plant cannot carry out active transport of water, and it wilts. If the chloroplasts stop working, the plant cannot photosynthesise and will starve'). Red flag: entry is generic ('the cell needs all its parts to work') without naming specific organelles or connecting to the phenomenon — teacher provides written prompt on the entry: 'Name two specific organelles. What does each one do? What happens to the sukuma wiki if that organelle fails?'

	anchoring phenomenon question — 'Why do the two microscopes show such different pictures, and what are all those internal structures doing to keep the sukuma wiki plant alive?'	questions still in STILL WONDERING: 'We have built our models, but these questions remain. What would we need to do to answer them? This is what scientists feel every day.' Finally, instruct: 'In your notebooks, write 2–3 sentences: How does your model connect to the sukuma wiki leaf that the learner in Kisumu observed wilting in the sun? Which organelles, if they stopped working, would cause the leaf to wilt and die?'		
Model Building Phase 📹 **VIDEO:** Cells and their Organelles - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 8 minutes, learners write a 'Before and After Science Story' directly on a label affixed to the base of their 3D model. The label has two columns: BEFORE THIS LESSON SEQUENCE (what I thought a cell was — from my Lesson 1 initial model) and AFTER BUILDING THIS MODEL (what I now know, with evidence). Learners compare their current 3D model to the sketch they drew in Lesson 1 (retrieved from their notebooks), identify the biggest change in their understanding, and write it in the AFTER column. They also add one final tag to their model — a 'PHENOMENON CONNECTION TAG' — which answers in one sentence: 'This model explains the wilting sukuma wiki because…'	Distribute the 'Before and After' base labels. Say: 'Go back to your Lesson 1 notebook sketch — the one you drew before we looked at any microscopes. Compare it to the model in front of you. What is the biggest difference? That difference IS your learning.' [WAIT 15 seconds for learners to locate Lesson 1 sketches.] 'Write that difference in the AFTER column. Then write your PHENOMENON CONNECTION TAG — one sentence that explains how your model answers the question of why the sukuma wiki wilted.' Circulate for 5 minutes, reading labels. Cold-call 2 learners for a closing share-out: 'Mwende, share your phenomenon connection tag.' [WAIT 10 seconds.] 'Korir, what is the single biggest change between your Lesson 1 sketch and your model today?' [WAIT 12 seconds.] Close with: 'Next lesson — our final lesson — we will use everything we have built, observed, and modelled to answer the driving question completely and present our understanding. Your model will be your evidence.'	Cumulative Model Revision with Before/After Comparison: by placing the Lesson 1 sketch and the Lesson 7 3D model side by side, learners perform an explicit cognitive comparison that reveals the depth and direction of their conceptual change. The PHENOMENON CONNECTION TAG is a one-sentence scientific claim — the most distilled form of sensemaking — requiring learners to synthesise seven lessons of evidence into a single coherent statement. This mirrors how scientists summarise findings in an abstract or conclusion.	Teacher reads PHENOMENON CONNECTION TAGS during circulation and final share-out. Observable indicator of full understanding: the tag contains a causal mechanism linking cell organelles to the phenomenon (e.g., 'This model explains the wilting sukuma wiki because when a cell loses water from its vacuole and the mitochondria can no longer supply energy for active processes, the cell loses turgor pressure and the leaf droops — and eventually dies'). Red flag: tag states only 'because cells are the basic unit of life' without a mechanism — teacher asks orally: 'Which specific organelle is involved in keeping the leaf firm? What does it hold? What happens when it empties?' Award verbal praise specifically for the quality of reasoning, not the aesthetic quality of the model, to reinforce that scientific accuracy is the goal.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal: 1. MODELLING QUALITY: Did learners' material choices reflect genuine understanding of organelle structure and function, or were choices primarily aesthetic? Which organelles were most frequently misrepresented, and what does that tell you about gaps in Lessons 3–5? 2. JUSTIFICATION DEPTH: During your circulation, how many learners achieved a 'Structural + Functional + Phenomenon-linked' justification (the highest level on your checklist)? Which learners are still at 'Structural only'? Plan a targeted question for Lesson 8 for those learners. 3. DQB CURATION: Were there questions on the DQB that no learner's blue sticker covered? Those are the questions to prioritise in the Lesson 8 driving question resolution discussion. 4. PHENOMENON CONNECTION: In the notebook entries and PHENOMENON CONNECTION TAGS, did learners consistently link organelle function to the wilting sukuma wiki? If learners' tags were generic, consider using two strong examples from today as mentor texts to open Lesson 8. 5. EQUITY: Was the materials distribution equitable across gender and socioeconomic groups? Did any group feel their materials were inadequate? Note this for future project lessons — pre-sorting materials into equal kits avoids this issue. 6. CREATIVITY AND IMAGINATION COMPETENCY: The KICD mandates Creativity and Imagination as a core competency for this sub-strand. Did the model-building task genuinely elicit creative thinking, or did learners copy each other's material choices? If copying occurred, consider assigning plant vs. animal cell to alternate learners from the start, so no two adjacent models are identical.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed (via ARES platform review) the annotated electron microscope photomicrographs of the sukuma wiki plant cell and an animal cell — revisiting the folded membranes, dark oval bodies, and large fluid-filled vacuole that puzzled the Kisumu learner in the original phenomenon. They also observed the physical properties of locally available materials (bottle tops, polythene bags, maize grains, bean seeds, clay, string, cardboard) at the Materials Table, evaluating which material best captures each organelle's structure and function.
What did I learn?	Learners demonstrated consolidated knowledge of the names, spatial arrangements, and functions of major plant and animal cell organelles as revealed under the electron microscope. By constructing, labelling, and justifying a 3D model, they showed understanding of: (a) the structural differences between plant and animal cells (cell wall, chloroplasts, large central vacuole present only in plant cells); (b) the function of each organelle in keeping the cell — and therefore the whole organism — alive; (c) the levels of cell organisation from organelle to organism. The DQB revisit revealed which aspects of the anchoring phenomenon learners can now explain and which remain open for Lesson 8.
How does this explain the phenomenon?	Learners explained, through their models and PHENOMENON CONNECTION TAGS, how the organelles visible in the electron microscope photomicrograph of the sukuma wiki leaf cell collectively keep the plant alive — and why the leaf wilts and dies when those organelles can no longer function. They articulated the causal chain: chloroplasts capture light energy → mitochondria release energy for cell processes → vacuole maintains turgor pressure → cell wall provides structural support → without these working together, the sukuma wiki leaf wilts, dries, and dies. This explanation directly answers the anchoring phenomenon and advances the class toward the full driving question resolution in Lesson 8.

LESSON 8 (40 minutes): Driving Question Board Synthesis and Final Assessment: What Keeps the Sukuma Wiki Leaf Alive?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and publicly demonstrate understanding of cell structure, organelle function, and levels of organisation by synthesising all prior learning into a coherent explanation of the anchoring phenomenon, and to transition outstanding questions toward Sub-Strand 1.3 Nutrition in Animals.
Knowledge	Learners demonstrate that: (i) plant and animal cells contain organelles visible under the electron microscope, each with a specific function that sustains life; (ii) plant cells uniquely possess a cell wall, chloroplasts, and a large central vacuole; (iii) cells are organised into tissues, organs, organ systems, and whole organisms such as the sukuma wiki plant and the human body in Nairobi.
Skills	Learners practise: peer evaluation through structured gallery walk feedback; scientific writing by constructing an explanatory paragraph linking cell structure to the survival of the sukuma wiki leaf; drawing and labelling a plant cell from a photomicrograph with accuracy and appropriate scale annotation.
Attitudes	Learners appreciate that scientific models are provisional and improvable, value peer critique as a tool for refining understanding, and show curiosity about how cell biology connects to everyday nutrition, farming, and health in Kenya.
Key Inquiry Question	What makes up an organism?
Purpose in Storyline	Lesson 8 is the capstone of the Sub-Strand 1.2 storyline. Learners began by wondering why a sukuma wiki leaf wilts and dies, and why a light microscope and electron microscope reveal such different levels of detail. Over seven lessons they built knowledge piece by piece. This lesson closes the explanatory loop: learners publicly display their 3D models, critique each other constructively, return as a class to the original wilting phenomenon, and together write a full evidence-based explanation. A short performance task checks individual mastery before the DQB is formally closed and outstanding questions are handed forward to Sub-Strand 1.3.
Safety Notes	No hazardous chemicals or live specimens are used in this lesson. Scissors or pins used to attach labels or sticky notes should be handled carefully. If learners move furniture for the gallery walk, they should do so calmly to avoid injury. Remind learners not to touch the lenses of any microscope or photomicrograph display equipment without permission.

B. LESSON OVERVIEW
Learners open the lesson by arranging their completed 3D cell models for a structured gallery walk in which they leave peer-feedback sticky notes on two other groups models. The class then reconvenes and collectively builds a final written explanation of the anchoring phenomenon — why the sukuma wiki leaf stays alive and what happens at the cellular level when it wilts and dies. Each learner individually completes a short performance task: drawing and labelling a plant cell from a photomicrograph and writing an explanatory paragraph. The lesson closes with a ceremonial DQB review in which answered questions are celebrated, remaining open questions are flagged, and learners identify which of those questions will be answered in Sub-Strand 1.3 Nutrition in Animals. The lesson is designed to last exactly 40 minutes across five instructional phases.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** [Cells and their Organelles - Example 1] Source: CK-12 🔍 [Search ARES for similar videos] 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Duration: 5 minutes. Learners enter to find their 3D cell models already displayed on desks or a central table in a gallery arrangement. Before the gallery walk begins, the teacher projects the original anchoring phenomenon image: a wilting sukuma wiki leaf beside a fresh sukuma wiki leaf photomicrograph. Learners spend 2 minutes silently re-reading the single question written on the DQB from Lesson 1: 'What are those tiny box-like compartments, what is inside them, and how does what is inside them keep the sukuma wiki leaf alive?' Each learner writes in their exercise book one sentence predicting: 'By the end of this lesson I will be able to explain...' This activates metacognitive awareness of how far their understanding has grown. Learners then receive three sticky notes each — one green (strength), one yellow (question or suggestion), one pink (connection to the phenomenon) — to use during the gallery walk in the next phase.	Teacher displays the wilting sukuma wiki image and says: 'Before we celebrate what you have built, I want you to travel back in your memory to Lesson 1. Look at this image. Look at this question on the DQB. Silently, write one sentence — just one — that begins: By the end of this lesson I will be able to explain...' WAIT TIME: 90 seconds of complete silence. Teacher then says: 'Hold that sentence in your mind. It is your personal target for the next 35 minutes. Take your three sticky notes. You will use them during our gallery walk — green for a strength you genuinely notice, yellow for a question or one thing the group could improve, and pink for a direct connection to our sukuma wiki phenomenon. Be specific. Vague praise teaches nobody anything.'	Metacognitive prediction prompt: learners articulate their own learning target before synthesis begins, connecting back to the original anchoring phenomenon. This primes selective attention during the gallery walk and performance task.	Teacher circulates and reads 3 to 4 learner prediction sentences over their shoulders. Checks whether learners are framing targets around cell structure, organelle function, and levels of organisation — the three pillars of the sub-strand. Mentally notes any learner who writes a vague sentence to follow up with a probing question during the gallery walk.
Observe Phase 📹 **VIDEO:** [Cells and their Organelles - Example 1] Source: CK-12 🔍 [Search ARES for similar videos] 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Duration: 8 minutes. Groups rotate around the room in a structured gallery walk. Each group spends approximately 90 seconds at each of two assigned models (models belonging to groups they did not build). Learners examine the 3D model, read the organelle function tags, and each place one sticky note: one learner places the green note, one places the yellow note, one places the pink note. Groups rotate on the teacher's signal (a clap or bell). After visiting two models, groups return to their own model and read the sticky notes	Teacher says: 'You have 90 seconds at each model. Use your eyes, use the tags, use what you know. Your feedback must be specific enough that the group can act on it. If you write Good model, you have wasted your sticky note and their time. Instead write: The chloroplast is labelled but the tag does not explain that it traps sunlight for photosynthesis in the sukuma wiki leaf — that would be a yellow note.' Teacher circulates and models the behaviour by thinking aloud at one model: 'I can see the mitochondria are placed	Structured gallery walk with colour-coded sticky notes ensures that peer feedback is categorised as evaluative (green), interrogative (yellow), or connective (pink), training learners to distinguish types of scientific commentary. Reading feedback on their own model prepares groups for the collaborative explanation phase.	Teacher photographs two or three models for display during the explanation phase. Teacher notices which models are missing key plant-cell-specific structures (cell wall, chloroplast, large central vacuole) or have incorrect function tags — these become targeted discussion points in the explain phase. Teacher checks whether pink sticky notes genuinely connect to the phenomenon or are generic.

	left for them. Each group has 1 minute to discuss: 'Which feedback will we use to improve our explanation in the next phase?' Learners record the single most useful piece of feedback in their exercise books. The teacher also walks the gallery, pausing at each model for 20 to 30 seconds to observe the quality of labelling and function tags.	near the cell membrane — I wonder why that choice was made. That is a genuine question worth a yellow note.' After rotation teacher says: 'Back to your own model. Read what your peers wrote. One minute. Choose the single most useful piece of feedback and write it in your book. You will use it in the explanation phase.' WAIT TIME: 60 seconds silent reading at own model.		
Explain Phase 📹 **VIDEO:** Cells and their Organelles - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Duration: 15 minutes. Part 1 — Collaborative class explanation (7 minutes): Class reconvenes. Teacher projects the original anchoring phenomenon image alongside the DQB. Using a structured whole-class co-construction strategy, the teacher builds a paragraph on the board sentence by sentence, cold-calling individual learners to contribute. The paragraph answers the question: 'Why does the sukuma wiki leaf stay alive, and what happens inside its cells when it wilts and dries?' Learners must use evidence from their 7-lesson journey: organelle names and functions, plant-cell-specific structures, and at least one level of organisation. Learners copy the completed paragraph into their books. Part 2 — Individual performance task (8 minutes): Teacher distributes a printed or projected photomicrograph of a plant cell (sukuma wiki or spinach cell electron micrograph, clearly showing nucleus, chloroplasts, cell wall, vacuole, mitochondria, cell membrane, and endoplasmic reticulum). Each learner individually: (i) draws and labels the cell, identifying at least six organelles; (ii) writes a paragraph	For Part 1 teacher says: 'We built this question together in Lesson 1. Now we answer it together. I will write the opening clause on the board: The sukuma wiki leaf stays alive because its cells contain specialised structures called... — who will complete this sentence with precise scientific language?' Cold-calls a learner. After each contribution teacher asks the class: 'Does this sentence use evidence? Does it connect to what we observed in the photomicrographs? Thumbs up if yes, thumbs sideways if it needs more precision.' Teacher refines contributions to model scientific writing without dismissing learner language. Teacher says: 'Notice we are writing the way scientists write — claim, evidence, reason.' For Part 2 teacher says: 'Now I need to see what YOU individually understand. This is your moment. No group support. The photomicrograph is your evidence. Your diagram must be drawn from the image, not from memory alone. Your paragraph must explain — not just describe. There is a difference: describing says what is there; explaining says what it does and why that matters for life.' WAIT TIME during performance task:	Co-constructed explanatory paragraph models scientific writing for the class before learners attempt individual writing, scaffolding the transition from collaborative to independent performance. The performance task requires both representational competence (labelled diagram from a photomicrograph) and explanatory competence (paragraph linking structure to function to survival), directly assessing the two highest-demand SLOs. The probing question during circulation ('what would happen if this structure stopped working') pushes causal reasoning beyond recall.	Teacher assesses the co-constructed paragraph for class-level accuracy before moving to Part 2 — correcting any misconceptions in the paragraph before learners consolidate them in individual writing. During individual performance task teacher uses a rapid mental checklist: Does the diagram include at least six labelled organelles? Does the paragraph include at least one plant-cell-specific structure, at least one function linked to survival, and at least one reference to a level of organisation beyond the cell? Collected performance tasks are reviewed after the lesson using the same checklist as a written formative record.

	(minimum 4 sentences) explaining how the structures they have labelled work together to keep the plant cell — and therefore the whole sukuma wiki plant — alive. Teacher circulates and asks targeted probing questions to individual learners. Learners submit the performance task at the end of the lesson for teacher review.	teacher does not speak for the first 3 minutes, allowing learners to engage with the image independently. Teacher then circulates and asks individual learners: 'Point to one structure in your diagram and tell me what would happen to this cell if that structure stopped working.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Cells and their Organelles - Example 1] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Cell Transport (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Duration: 7 minutes. Teacher brings out the original DQB from Lesson 1 (or a photographed/reprinted version). Learners read all the questions that were posted in Lesson 1 under 'What we wonder.' In pairs, learners sort the questions into three columns on a shared sheet: (A) Fully answered — we can explain this now; (B) Partly answered — we have some evidence but questions remain; (C) Not yet answered — we will need to investigate this further. Each pair reports one question from column C. Teacher records column C questions on a new section of the board labelled: 'Questions for Sub-Strand 1.3 and beyond.' Teacher makes explicit links: for example, questions about how the sukuma wiki plant makes food from sunlight (photosynthesis) connect to Sub-Strand 1.3 Nutrition in Plants; questions about how the human digestive system breaks down sukuma wiki so the body can use its nutrients connect to Sub-Strand 1.3 Nutrition in Animals. Teacher highlights that unanswered questions are not failures — they are the engine of scientific inquiry. Teacher says: 'Every scientist in Kenya — at KEMRI, at the	Teacher holds up the original DQB and says: 'This board was created in Lesson 1 when you were still puzzled by what you saw under the microscope. Look at it now with the eyes of someone who has built a 3D model, compared plant and animal cells, and just written a scientific explanation. Which of these questions can you now answer with confidence?' WAIT TIME: 60 seconds for pairs to read and sort silently. Teacher cold-calls three pairs for column A answers and affirms with evidence: 'Yes — and the evidence for that came from which lesson?' For column C teacher says: 'These unanswered questions are gifts. Write one of them at the top of the next clean page in your exercise book. It is the first question of your next investigation.' Teacher explicitly names the transition: 'When we begin Sub-Strand 1.3 Nutrition in Animals, you will find that the structures inside cells — especially the mitochondria — are central to how your body releases energy from the sukuma wiki you eat. Your question is already waiting for its answer.'	Tripartite DQB sort (fully answered, partly answered, not yet answered) makes metacognitive progress visible and communal, validating the learning journey while normalising scientific incompleteness. Explicit naming of Kenyan research institutions contextualises inquiry as a living, local practice. The transition statement provides a narrative bridge to Sub-Strand 1.3 so learners experience curriculum as a coherent story rather than disconnected topics.	Teacher listens to pair discussions during the sort to check whether learners are accurately judging what they do and do not yet understand — overconfidence (placing unanswered questions in column A) or underconfidence (placing answered questions in column C) both signal areas needing follow-up. Teacher notes which column C questions recur across multiple pairs, as these reveal persistent gaps to address at the start of Sub-Strand 1.3.

	University of Nairobi, at ICRAF in Gigiri — works on questions that are not yet answered. You are doing exactly what scientists do.'			
Model Building Phase 📹 **VIDEO:** Cells and their Organelles - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Cell Transport (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Duration: 5 minutes. This phase serves as a reflective closure on the Lesson 7 model-building activity rather than new construction. Groups briefly revisit their own 3D model in light of: (i) the sticky-note feedback received during the gallery walk; (ii) the co-constructed class explanation from the explain phase; (iii) any corrections or additions they now recognise as needed. Each group writes on a card placed beside their model: 'If we were to rebuild this model tomorrow, one thing we would change or add is...' and 'The most important organelle in this model for explaining why the sukuma wiki leaf stays alive is... because...' Groups share their cards aloud in a 60-second round-robin. Teacher collects the cards as an artefact of reflective practice. Teacher closes with a summary statement connecting the model to the phenomenon and to the learner as a citizen: 'The sukuma wiki plant on a farm in Kisii, the kale at Wakulima Market in Nairobi, the greens on your plate tonight — all of them alive because of the organelles you have just labelled, modelled, and explained. That knowledge belongs to you now.'	Teacher says: 'You have 2 minutes as a group. Look at your model. Look at the sticky notes. Look at the paragraph we wrote together. One thing you would change, and the single most important organelle for the phenomenon. Write it on your card.' WAIT TIME: 2 minutes of group discussion. Teacher then says: 'Round robin — each group, 10 seconds, read your most important organelle and one reason. Go.' Teacher listens and notes if any group identifies an organelle that is incorrect or a reason that is incomplete — offers a brief, affirming correction: 'The chloroplast is an excellent choice — can you add what would happen to the leaf if the chloroplasts stopped functioning?' Teacher closes lesson with the contextualising statement above, delivered slowly and with deliberate reference to specific Kenyan places. Teacher reminds learners to submit their performance task drawings and paragraphs before leaving the classroom.	The reflective model card task transforms the 3D model from a finished product into an evolving scientific representation, reinforcing the CBE principle that models are tools for thinking rather than endpoints. The round-robin sharing ensures every group contributes a public closing statement, consolidating the social dimension of sense-making. The teacher closing statement anchors cellular biology in the lived environment of Kenyan learners, fulfilling the sub-strand goal of connecting microscopic science to macroscopic life.	Model cards collected by teacher provide a written record of reflective self-assessment and are reviewed alongside performance task submissions to build a holistic picture of each group and individual learner. Teacher checks whether the 'most important organelle' choices and justifications reflect accurate functional reasoning. Round-robin responses allow a final whole-class sweep for persistent misconceptions before the sub-strand ends.

D. TEACHER REFLECTION

After the lesson, the teacher should consider the following questions: (1) Did the gallery walk generate genuinely specific peer feedback, or did sticky notes remain vague — and if so, what scaffolding would improve the quality of peer evaluation next time? (2) Did the co-constructed class paragraph accurately capture all three pillars — organelle names and functions, plant-cell-specific structures, and levels of organisation — or did one pillar dominate while others were thin? (3) When

reviewing the individual performance tasks, which organelles were most frequently mislabelled or omitted from diagrams, and which explanatory connections (structure to function to survival) were weakest in the paragraphs? These gaps should be noted as entry points for Sub-Strand 1.3. (4) Were the DQB column C questions substantive and genuinely connected to Sub-Strand 1.3 content, indicating that learners are thinking beyond the current sub-strand? (5) Did the lesson feel rushed at any phase — in particular, did the performance task receive sufficient uninterrupted individual time, or did earlier phases run long? Adjust timing boundaries for future delivery. (6) Were Kenyan contexts — Kisumu, Wakulima Market, KEMRI, ICRAF, sukuma wiki farming in Kisii — woven naturally into teacher talk, or did they feel inserted? Authentic contextualisation strengthens the CBE value of connecting learning to the learner's own environment.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In Lesson 8 we observed the 3D cell models built by all groups displayed in a gallery, the original DQB from Lesson 1 with all the questions generated at the start of the sub-strand, and a photomicrograph of a plant cell used as the basis of the individual performance task.
What did I learn?	We learned that a sukuma wiki leaf stays alive because its cells contain specialised organelles — including chloroplasts that capture sunlight, mitochondria that release energy, a nucleus that holds genetic instructions, a cell wall that provides structural support, and a large central vacuole that maintains turgor — and that these cells are organised into tissues, organs, and organ systems that make up the whole plant. We also confirmed that plant cells differ from animal cells in possessing a cell wall, chloroplasts, and a large central vacuole.
How does this explain the phenomenon?	We explained why the sukuma wiki leaf wilts and dies when left in the sun by connecting the loss of water from the central vacuole to loss of turgor, the breakdown of chloroplasts to the failure of photosynthesis, and the cessation of mitochondrial activity to the inability of cells to release the energy needed for all life processes — showing how the death of organelles causes the death of cells, then tissues, then the whole leaf.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.